

A *MSTN*^{Del273C} mutation with *FGF5* knockout sheep by CRISPR/Cas9 promotes skeletal muscle myofiber hyperplasia

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Abstract

Mutations in the well-known Myostatin (*MSTN*) produce a “double-muscle” phenotype, which makes it commercially invaluable for improving livestock meat production and providing high-quality protein for humans. However, mutations at different loci of the *MSTN* often produce a variety of different phenotypes. In the current study, we increased the delivery ratio of Cas9 mRNA to sgRNA from the traditional 1:2 to 1:10, which improves the efficiency of the homozygous mutation of biallelic gene. Here, a *MSTN*^{Del273C} mutation with *FGF5* knockout sheep, in which the *MSTN* and *FGF5* dual-gene biallelic homozygous mutations were produced via the deletion of 3-base pairs of AGC in the third exon of *MSTN*, resulting in cysteine-depleted at amino acid position 273, and the *FGF5* double allele mutation led to inactivation of *FGF5* gene. The *MSTN*^{Del273C} mutation with *FGF5* knockout sheep highlights a dominant “double-muscle” phenotype, which can be stably inherited. Both F0 and F1 generation mutants highlight the excellent trait of high-yield meat with a smaller cross-sectional area and higher number of muscle fibers per unit area. Mechanistically, the *MSTN*^{Del273C} mutation with *FGF5* knockout mediated the activation of *FOSL1* via the MEK-ERK-FOSL1 axis. The activated *FOSL1* promotes skeletal muscle satellite cell proliferation and inhibits myogenic differentiation by inhibiting the expression of MyoD1, and resulting in smaller myotubes. In addition, activated ERK1/2 may inhibit the secondary fusion of myotubes by Ca²⁺-dependent CaMKII activation pathway, leading to myoblasts fusion to form smaller myotubes.

eLife assessment

The authors present a **useful** analysis of the phenotype of sheep in which the muscle developmental regulator myostatin has been mutated in a *FGF5* knockout background. The goal was to produce sheep with a "double-muscled" phenotype, yet the genetically engineered sheep exhibited meat with a smaller cross-sectional area and higher number of muscle fibers. The work extends the extensive body of knowledge already published in this area. The authors provide evidence using in vitro experiments that *Fosl1* regulates myogenesis, but the strength of evidence relating to the muscle phenotype and underlying cellular and molecular mechanism remains **incomplete**.

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1 Introduction

Myostatin (*MSTN*) has been well-known as a negative regulator of muscle growth and development. Its mutation produces a "double-muscle" phenotype, which shows its inestimable commercial value in improving meat production of livestock and poultry, and providing high-quality protein for humans (Fan *et al.*, 2022; Chen *et al.*, 2021b). Due to its role in promoting muscle atrophy and cachexia, *MSTN* has been recognized as a promising therapeutic target to offset the loss of muscle mass (Lee, 2021; Baig *et al.*, 2022; Wijaya *et al.*, 2022).

MSTN is highly conserved in mammals, and mutations in the *MSTN* gene, either artificially or naturally, will result in increased skeletal muscle weight and produce a "double-muscle" phenotype, which has been reported in many species, including cattle, sheep, and pigs, rabbits, and humans (Grisolia *et al.*, 2009; Dilger *et al.*, 2010; Kambadur *et al.*, 1997). However, mutations at different loci of the *MSTN* often produce variety of different phenotypes, and its molecular mechanism of skeletal muscle growth and development remains controversial (Hanset and Michaux, 1985; Grobet *et al.*, 1997; Wegner *et al.*, 2000; Kambadur *et al.*, 1997; Marchitelli *et al.*, 2003). More than 77 natural mutation sites of *MSTN* have been reported in various sheep breeds, most of these mutations were found to be located in the non-coding regions, and did not affect *MSTN* activity (Kijas *et al.*, 2007; Sjakste *et al.*, 2011; Han *et al.*, 2013; Dehnavi *et al.*, 2012). In addition to introns, it is still possible that mutations in regulatory regions and exons may not affect the sheep phenotypes (Pothuraju *et al.*, 2015; Kijas *et al.*, 2007; Boman and Vage, 2009; Boman *et al.*, 2009).

Fibroblast growth factor 5 (*FGF5*) belongs to the fibroblast growth factor (FGF) family and is a secretory signaling protein. *FGF5* played an inhibitory effect on mouse hair growth (Hebert *et al.*, 1994), and its natural mutation can lead to a significant increase in hair growth in angora mice (Sundberg *et al.*, 1997). Subsequent studies have also successively confirmed the inhibitory effect of *FGF5* on mammalian hair growth and is recognized to be a negative regulator of hair growth (Kehler *et al.*, 2007; Dierks *et al.*, 2013; Yoshizawa *et al.*, 2015; Legrand *et al.*, 2014; Higgins *et al.*, 2014).

In this study, to increase both meat and wool production, we first produced the *MSTN* and *FGF5* dual-gene biallelic homozygous mutations sheep by the increased delivery ratio of Cas9 mRNA to sgRNA targeting *MSTN* and *FGF5*. The *MSTN*^{Del273C} mutation with *FGF5* knockout sheep highlights a dominant "double-muscle" phenotype by decreasing the muscle fiber cross-sectional area and

increasing the number of muscle fibers per unit area. Then, we used the *MSTN* and *FGF5* dual-gene biallelic homozygous mutations sheep to unravel the molecular mechanism of the “double-muscle” phenotype and myofiber hyperplasia.

2 Materials and Methods

2.1 Production of Cas9 mRNA and sgRNA

The Cas9 and U6-sgRNA co-expression vector backbones pX330 were purchased from Addgene (plasmid ID: 42230 and 48138). Sheep *MSTN* and *FGF5* sgRNAs were designed using CRISPR Design Tool (<http://tools.genome-engineering.org>). The *MSTN* sgRNA (GACATCTTTGTAGGAGTACAGCAA) and *FGF5* sgRNA

(AGGTTCCCTTTCCGCACCT) were used in this study. Two complementary guide sequence oligos were synthesized, annealed, and cloned into the pX330 backbone vector to form the functional co-expression plasmids. T7 promoter was linked to the 5' ends of the Cas9 coding region and *MSTN*/*FGF5* sgRNA template by PCR amplification from the pX330-*MSTN*/*FGF5* plasmid constructed as described above. Then, these PCR products, as transcription templates, were purified using E.Z.N.A. Cycle Pure Kit (Omega Bio-Tek). *MSTN* sgRNA and *FGF5* sgRNA were prepared by in vitro transcription (IVT) using the MEGAshortscript T7 Kit (Life Technologies). Cas9 mRNA was transcribed with the m7G(5')ppp(5') G cap on its 5' terminal and poly (A) tail on its 3' terminal using the mMESSAGE mMACHINE T7 Ultra Kit (Life Technologies). Both the Cas9 mRNA and the sgRNAs were purified using the MEGAclean Kit (Life Technologies) and eluted in RNase-free water.

2.2 Microinjection and embryo transfer

The procedure for the efficient production of pronuclear embryos has been described previously (Li *et al.*, 2016). Briefly, Cas9 mRNA (1000 ng/μL) and sgRNAs (200 ng/μL) were mixed and injected into the transferable embryos in which the zona pellucida was clear, cytoplasm was uniform, and pronucleus was visible using a FemtoJet microinjector (Eppendorf). Following microinjection, three to five embryos were transplanted into the oviduct of each recipient within 1 h after starting the laparotomy operation. Pregnancy was confirmed by transabdominal ultrasound scanning on the 60th day after embryo transfer.

2.3 Tissue sample collection and preparation

Gluteus medius and longissimus dorsi were harvested from WT and *MSTN*^{Del273C} mutation with *FGF5* knockout (MF^{-/-}) sheep, and three WT sheep and four MF^{-/-} F1 generation sheep (half-sib) were used for feeding and slaughter. All sheep were female and are slaughtered at 12-month-old. All samples were immediately frozen in liquid nitrogen and then stored at -80°C until analysis. All sheep are raised by the national feeding standard NT/T815-2004. All procedures performed for this study were consistent with the National Research Council Guide for the Care and Use of Laboratory Animals. All experimental animal protocols in this study were approved and performed following the requirements of the Animal Care and Use Committee at China Agricultural University (AW02012202-1-3). All surgeries were performed under sodium pentobarbital anesthesia, and all efforts were made to minimize any suffering experienced by the animals used in this study.

2.4 H&E staining and morphological analysis of muscle fibers

Fresh muscle tissue samples were fixed, dehydrated, embedded and frozen sectioned, respectively. Next, they were sequentially stained with hematoxylin and eosin, then dehydrated with gradient ethanol and transparentized with xylene, and finally sealed with neutral resin to make tissue

sections. The images from at least five random fields were captured with an inverted microscope. The Image J software was used to segment, count and calculate the area of each muscle fiber cell. The number of fiber cell in a fixed area size was calculated to estimate the number of fiber cell per unit area.

2.5 Cell isolation, culture, and transfection

Sheep skeletal muscle satellite cells were isolated and cultured as previously described (Chen *et al.*, 2021a [DOI](#)). In brief, the muscle tissues of the hind limbs from 3-month-old sheep fetuses were cut into small pieces, digested sequentially with 0.2% collagenase type II (Gibco, Grand Island, NY) and 0.25% trypsin (Gibco, Grand Island, NY). The cell suspension was successively filtered through 100, 200 and 400 mesh cell sieves. After this, the cells were resuspended in growth medium (GM) containing DMEM/F12 (Gibco, Grand Island, NY) with 20% fetal bovine serum (FBS, Gibco) and 1% penicillin-streptomycin liquid (Gibco, Grand Island, NY), and cultured for 2-3 times with differential adhesion. To induce differentiation, the cells were cultured to 70% confluence in GM, and followed by an exchange to differentiation medium (DM) containing DMEM high glucose (Gibco, Grand Island, NY) with 2% horse serum (HS, Gibco) and 1% penicillin-streptomycin. To produce viral solution for over-expression of the target gene, it was subcloned into the XbaI and BamHI sites of the lentiviral vector by seamless cloning, and the primer sequences of gene cloning were listed in Table S1. HEK 293T cells were co-transfected with the envelope plasmid pMD2.G, the packaging plasmid psPAX2 and the target plasmid at a mass ratio of 1:2:4 to produce the virus. The siRNA were synthesized by Guangzhou RiboBio Co., Ltd, and the sequences were listed in Table S2. Then, the cells were infected with packaged lentivirus or transfected with siRNA using Lipofectamine 3000 (Invitrogen, USA) when they were cultured to 60%-70% confluence.

2.6 Total RNA isolation and real-time quantitative PCR (RT-qPCR)

The total RNA of tissues and cells was isolated using TRIzol reagent (Sangon Biotech, Shanghai, China) following the manufacturer's protocol. In short, after tissues or cells were lysed, chloroform was added to separate the organic and inorganic phases, followed by precipitation with isopropanol and ethanol in turn, and finally, the RNA was dissolved in DEPC water. Then, the first strand cDNA was prepared using PrimeScript II 1st Strand cDNA Synthesis Kit (Takara, Beijing, China).

qPCR was performed using 2× SYBR Green qPCR Mix (Low ROX) (Aidlab Biotechnologies, Beijing, Chian) in a Stratagene Mx3000P (Agilent Technologies, SUA). With GAPDH mRNA as endogenous control, the relative expression level of genes was calculated by the $2^{-\Delta\Delta C_t}$ method. All primers used were listed in Table S3.

2.7 Western blot

Tissue or cell samples were lysed in RIPA buffer (Solarbio, Beijing, China) supplemented with protease and phosphatase inhibitor cocktail (Beyotime, Beijing, China) for total protein extraction. Then, equal amounts of tissue or cell lysate were resolved by 10% SDS-PAGE and transferred onto PVDF membranes (Millipore, USA). The membranes were blocked with 5% BSA for 1h, incubated with primary antibody at 4°C overnight, then incubated with secondary antibody for 1h before detection. The fold change of protein was normalized to GAPDH for quantitative analysis by ImageJ software. The antibodies information was listed in Table S4.

2.8 5-Ethynyl-2'-deoxyuridine (EdU) assay

At 24 h after transfection, sheep skeletal muscle satellite cells were incubated at 37°C for 2 h in 96-well plates with 50 μM EdU (RiboBio, Guangzhou, China). Then, fixed the cells in 4% paraformaldehyde for 30 min and neutralized using 2 mg/mL glycine solution. The Apollo[®] staining solution which contains EdU was added and incubated at room temperature for 30 min in the dark to label the DNA in the synthesis stage, the nuclear was then counterstained with DAPI.

The number of EdU positive cells was counted from the images of five random fields obtained with an inverted fluorescence microscope at a magnification of 100×. EdU labeling index was expressed as the number of EdU-positive cell nuclei/total cell nuclei.

2.9 Cell counting kit-8 (CCK-8) and cell cycle detection

Skeletal muscle satellite cells were seeded in 96-well plates and cultured for appropriate time according to different experimental treatments. Then, 10 μ L CCK-8 solution was added to each well and incubated at 37°C in a 5% CO₂ incubator for 2 h, and then the absorbance at 450 nm was measured with a microplate reader. The cultured skeletal muscle satellite cells were digested with trypsin, centrifuged at 1000 g for 5 min to collect the cell pellet, washed once with ice-cold PBS, and then 1 mL of ice-cold 70% ethanol was added to fix the cells overnight at 4°C. The next day, the cells were washed with ice-cold PBS again, and the cells were incubated with 0.5 mL PI staining solution at 37°C for 30 min and collected by flow cytometry at low speed.

2.10 Immunofluorescence staining

Sheep skeletal muscle cells were fixed in 4% paraformaldehyde for 30 min, permeabilized in 0.1% Triton X-100 for 20 min and blocked with 5% normal goat serum for 30 min at room temperature, and then incubated with primary antibody at 4°C overnight. Next, the fluorescent secondary antibody was added and incubated at 37°C for 1 h in the dark, and the nuclear was then counterstained with DAPI. The immunofluorescence images from five random fields were captured with an inverted fluorescence microscope.

2.11 Chromatin Immunoprecipitation (ChIP)

The cells were fixed with 1% formaldehyde, then the cells were collected with a cell scraper and resuspended in cell lysis buffer (10 mM HEPES, 0.5% NP-40, 1.5 mM MgCl₂, 10 mM KCl, pH 7.9) containing protease inhibitor cocktail (Beyotime, Beijing, China) and incubated on ice to release the cytoplasm. Next, cell pellets were collected and resuspended in nuclear lysis buffer (50 mM Tris, 10 mM EDTA, 0.3% SDS, pH 8.0) containing protease inhibitor cocktail. After the DNA was fragmented by ultrasonication, the supernatant was collected. The samples were diluted ChIP dilution buffer (0.01% SDS, 1.1% Triton X-100, 1.2 mM EDTA, 16.7 mM Tris-HCl pH 8.0, 167 mM NaCl), then 5 μ g primary antibody was added and incubated overnight at 4°C with rotation. The next day, protein A/G magnetic beads were added to each sample and incubated at 4°C with rotation for 2h. Then, the magnetic beads were respectively washed once with low-salt wash buffer (0.1% SDS, 1% Triton X-100, 2 mM EDTA, 20 mM Tris-HCl pH 8.0, 150 mM NaCl), high-salt wash buffer (0.1% SDS, 1% Triton X-100, 2 mM EDTA, 500mM NaCl, 20 mM Tris-HCl pH 8.0), LiCl wash buffer (0.25 M LiCl, 1% NP-40, 1% sodium deoxycholate, 1 mM EDTA, 10 mM Tris-HCl pH 8.0), and TE buffer (10 mM Tris-HCl pH 8.0, 1 mM EDTA) at 4°C. Next, ChIP elution buffer (1% SDS, 100 mM NaHCO₃) containing proteinase K was added to each sample, then incubated at 62°C overnight, and the DNA was finally purified by a purification column.

2.12 RNA-seq

The rRNA was removed from each total RNA sample of the gluteus medius to construct a strand-specific transcriptome sequencing library, and the Illumina Novaseq 6000 sequencing platform was used to perform high-throughput sequencing with a paired-end read length of 150 bp. Raw data were transformed into clean reads by removing reads containing adapter, ploy-N and low-quality reads from raw data. At the same time, Q20, Q30, GC-content, and sequence duplication levels of the clean data were calculated. The genome index was constructed using Hisat2 software and the clean reads were mapped to the sheep reference genome (*Oar Rambouillet v1.0*), the featureCounts software was used for expression quantification, and DESeq2 software was used for differential expression analysis based on P -value < 0.05 and $|\log_2 \text{Fold Change}| > 1$.

3 Results

3.1 Elevated molar ratio of Cas9/sgRNA can efficiently generate biallelic homozygous mutant sheep

The sgRNAs for targeting were designed in the third exon of the *MSTN* and *FGF5* genes, respectively (Figure 1A, B). Both *MSTN* and *FGF5* PCR products could be cleaved by T7E1 and the fragment sizes were also as expected, and grayscale analysis showed that the editing efficiency was 14.6% and 11.4%, respectively (Figure 1C), which indicates that the designed sgRNAs can achieve more efficient gene targeting. The microinjection was performed according to the molar ratio of Cas9 mRNA:sgRNAs (1:2, 1:10, and 1:15), respectively. The number of embryos injected, recipients of nuclear transfer, pregnancy, and alive lambs per group were listed in Table S5. The subsequent gene mutation detection showed that a total of 3 lambs were mutated in the *MSTN* and *FGF5* genes at a Cas9 mRNA:sgRNAs injection molar ratio of 1:2, with a gene editing mutation rate of 14.3% (4/28).

However, all 3 lambs were chimeric, that is, there were both mutant and wild-type after editing, and the biallelic mutation rate was 0% (Table S5, Figure 1D). Increasing the injection molar ratio of Cas9 mRNA:sgRNAs to 1:10 resulted in mutations in *MSTN* and *FGF5* genes of two lambs with a gene editing mutation rate of 18.2 % (4/22), which contributed to a significant ($P<0.05$) increase in the biallelic mutation rate (Table S5, Figure 1D). While the injection molar ratio of Cas9 mRNA: sgRNAs was continuously increased to 1:15, one lamb had a mutation, which was a biallelic mutation of *MSTN* gene, and the gene editing mutation rate was 7.14% (1/14) (Table S5, Figure 1D). These results indicate that increasing the delivery molar ratio of Cas9 mRNA to sgRNA from 1:2 to 1:10 can greatly improve the efficiency of biallelic mutation in sheep.

3.2 The *MSTN*^{Del273C} mutation with *FGF5* knockout sheep highlights a dominant “double-muscle” phenotype and muscle fiber hyperplasia

Among gene-edited sheep, a sheep with biallelic deletion of *MSTN* and biallelic mutation of *FGF5* aroused our great interest. Specifically, gene editing caused a deletion of 3-base pairs of AGC in the third exon of *MSTN* (Figure 1E, F), resulting in the deletion of cysteine at amino acid position 273 (*MSTN*^{Del273C}) (Figure 1E, F), which is highlighted by the “double-muscle” phenotype (Figure 2A, 2B). At the same time, a biallelic mutation in *FGF5* caused the knockout of *FGF5* gene and increased the density and length of hairs (Zhang *et al.*, 2020). Compared to WT sheep, the fiber cell number per unit area of gluteus medius and longissimus dorsi in MF^{-/-} sheep was significantly ($P<0.01$) increased (Figure 2C, D), and the cross-sectional area were smaller (Figure 2C, E-F). Similarly, the cross-sectional area of gluteus medius muscle fibers in the offspring generation MF^{+/-} sheep was also smaller (Figure 2G-H), and the number of muscle fiber cells per unit area was significantly increased ($P<0.0001$) (Figure 2G, I); the percentage of smaller muscle fiber area in MF^{+/-} sheep was significantly increased ($P<0.05$) (Figure 2G, J), these results was consistent with that in MF^{-/-} sheep. Interestingly, the *FGF5* knockout alone had no significant ($P>0.05$) effect on muscle fiber size (Figure S1A-D). Although the mRNA expression levels of *MSTN* and *FGF5* were significantly ($P<0.05$) reduced in MF^{+/-} sheep during early embryonic development (3-month-old) (Figure S2A), there was no significant differences in *MSTN* mRNA and protein expression levels at 12-month-old after birth (Figure S2B-D). In addition, there was no significant difference in the proportion of centrally nucleated myofibers ($P>0.05$) (Figure S2E), nor were there aberrant expression of some genes related to muscular

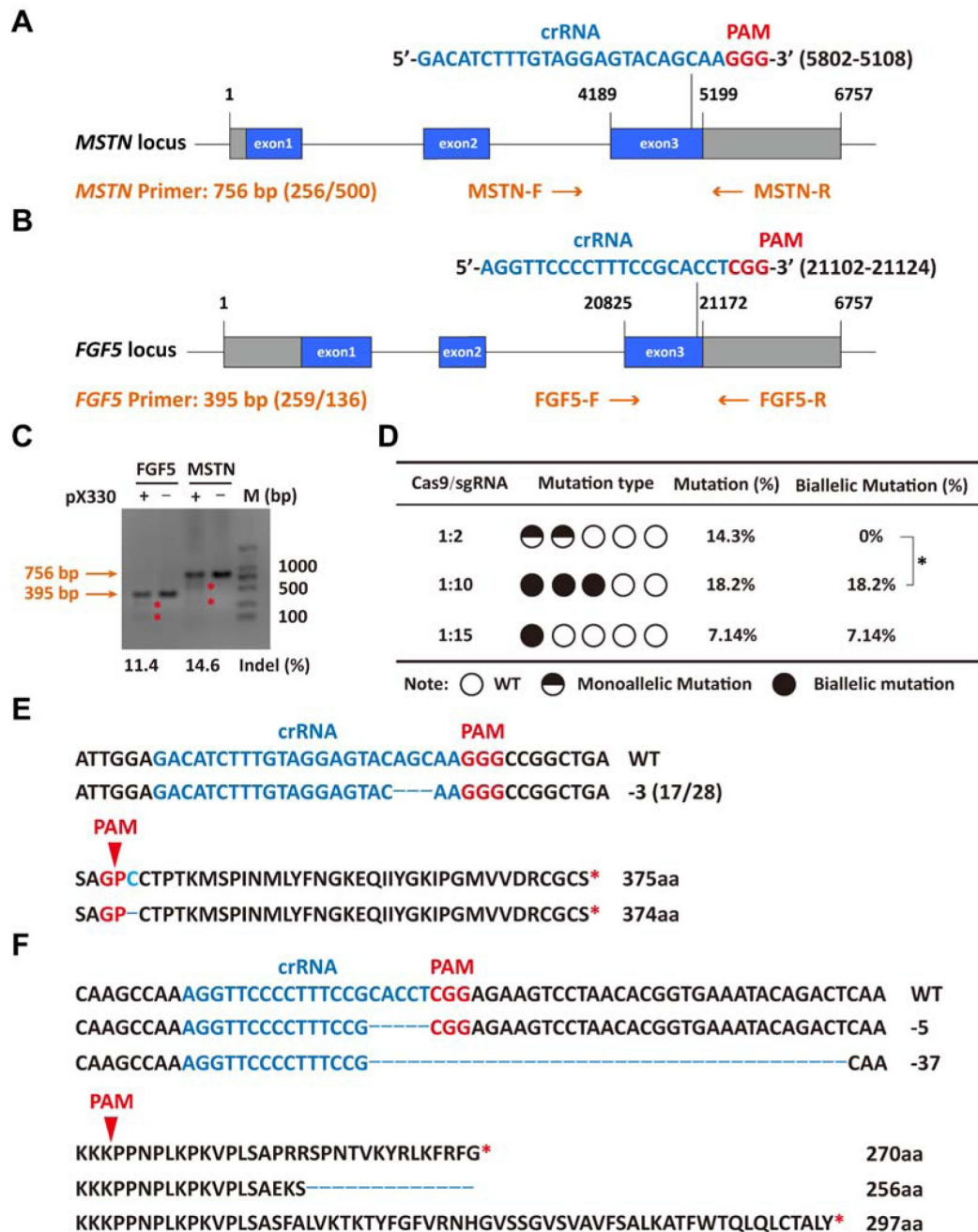


Figure 1

Efficient generation of sheep carrying biallelic mutations in dual gene via the CRISPR/Cas9 system

(A) Schematic of sgRNAs specific to exon 3 of the sheep MSTN locus. The crRNA sequences are highlighted in blue typeface and the PAM in red. (B) Schematic of sgRNAs specific to exon 3 of the sheep FGF5 locus. The crRNA sequences are highlighted in blue typeface and the PAM in red. (C) T7EI assay for sgRNAs of MSTN and FGF5 in sheep fetal fibroblasts. The cleavage bands are marked with an red asterisk (*) and the indel frequencies were calculated using the expected fragments. (D) Summary of the generation of sheep carrying biallelic mutations in dual gene via zygote injection of Cas9 mRNA/sgRNAs. Biallelic mutation rate was statistically analyzed using chi square test. * $P < 0.05$. (E) Analysis of genome sequence and amino acid sequence of MSTN-modified sheep. The location of sgRNA and PAM are highlighted in blue and red, respectively. The deletions are indicated by a dashed line (-). (F) Analysis of genome sequence and amino acid sequence of FGF5-modified sheep. The location of sgRNA and PAM are highlighted in blue and red, respectively. The deletions are indicated by a dashed line (-).

dystrophy and muscle atrophy (**Figure S2F**). These results indicate that *MSTN*^{Del273C} mutation with *FGF5* knockout produces muscle fiber hyperplasia instead of muscular dystrophy or muscle atrophy.

Further, although the muscle weight of different parts in WT and MF^{+/-} sheep has no significant difference (Table S6), the proportion of hind leg meat was significantly ($P < 0.05$) increased by 21.2% (Table S7), and the proportion of gluteus medius in the carcass of MF^{+/-} sheep was significantly ($P < 0.01$) increased by 26.3% compared to WT sheep (**Figure 2K**). In addition, there were no significant ($P > 0.05$) differences in pH, color, drip loss, cooking loss, shearing force, and amino acid content of the longissimus dorsi between WT and MF^{+/-} sheep (Table S8-10). All these results demonstrated that the *MSTN*^{Del273C} mutation with *FGF5* knockout sheep had well-developed hip muscles with smaller muscle fibers, which do not affect meat quality, and this phenotype may be dominated by *MSTN* gene.

3.3 The *MSTN*^{Del273C} mutation with *FGF5* knockout promotes skeletal muscle satellite cells proliferation and inhibits myogenic differentiation

The proliferation and differentiation of skeletal muscle satellite cells is a key step in muscle formation and development. The CCK-8 and EdU cell proliferation experiments showed that the proliferative rate of MF^{+/-} cells were highly significantly ($P < 0.01$) elevated (**Figure 3A**) with a significant ($P < 0.05$) increase in the rate of EdU-positive cells (**Figure 3B, C**) compared to WT cells. In addition, cell cycle detection showed a significant ($P < 0.01$) reduce in the proportion of G1 phase and a significant increase ($P < 0.05$) in the proportion of S phase in MF^{+/-} cells (**Figure 3D, E**). Meanwhile, the mRNA expression levels of the cell cycle marker genes CyclinB1, CDK4, Cyclin A1, Cyclin E1, and CDK2 were significantly increased ($P < 0.05$) (**Figure 3F**). These results suggest that the *MSTN*^{Del273C} mutation with *FGF5* knockout may promote cell proliferation by accelerating the cell cycle from G0/G1 phase to S phase.

The mRNA level of MyHC and the protein levels of MyoD1, MyoG, and MyHC (**Figure 3G-I**) were dramatically decreased ($P < 0.05$) after induced differentiation 2 days in MF^{+/-} cells, suggesting that the *MSTN*^{Del273C} mutation with *FGF5* inhibit myogenic differentiation. Meanwhile, the immunofluorescence staining of MyoG and MyHC in myotubes showed that myotube fusion index (**Figure 3J, K**), number of myotubes (**Figure 3J, L**), and number of nuclei per myotube (**Figure 3J, M**) were all highly significantly ($P < 0.01$) reduced after inducing differentiation for 2 days of MF^{+/-} cells compared to WT cells, as was the myotube diameter at the maximum measured (**Figure 3J, N**). In addition, the differentiation capacity and fusion ability of MF^{+/-} cells were consistently significantly lower than WT cells during the ongoing differentiation process, as was the diameter of fused myotubes (**Figure S3A-J**). The reduced expression of myogenic differentiation markers further confirmed that the *MSTN*^{Del273C} mutation with *FGF5* knockout consistently inhibits myogenic differentiation of skeletal muscle satellite cells (**Figure S3K-N**). Taken together, our results elucidated that the *MSTN*^{Del273C} mutation with *FGF5* knockout promotes proliferation and inhibits myogenic differentiation of skeletal muscle satellite cells, and induces a smaller myotube diameter of myotubes, which may explain the muscle fiber hyperplasia phenotype and the decreased cross-sectional area of muscle fibers in MF^{+/-} and MF^{+/-} sheep.

3.4 The *MSTN*^{Del273C} mutation with *FGF5* knockout contribute to muscle phenotype via MEK-ERK-FOSL1 axis

To elucidate the potential mechanism of the *MSTN*^{Del273C} mutation with *FGF5* knockout result in smaller muscle fiber cross-sectional area and myotube diameter, the RNA-seq was performed in gluteus medius. GO and KEGG enrichment analysis indicating that differentially expressed genes (DEGs) were significantly closely related to cell proliferation, myogenic differentiation, and muscle

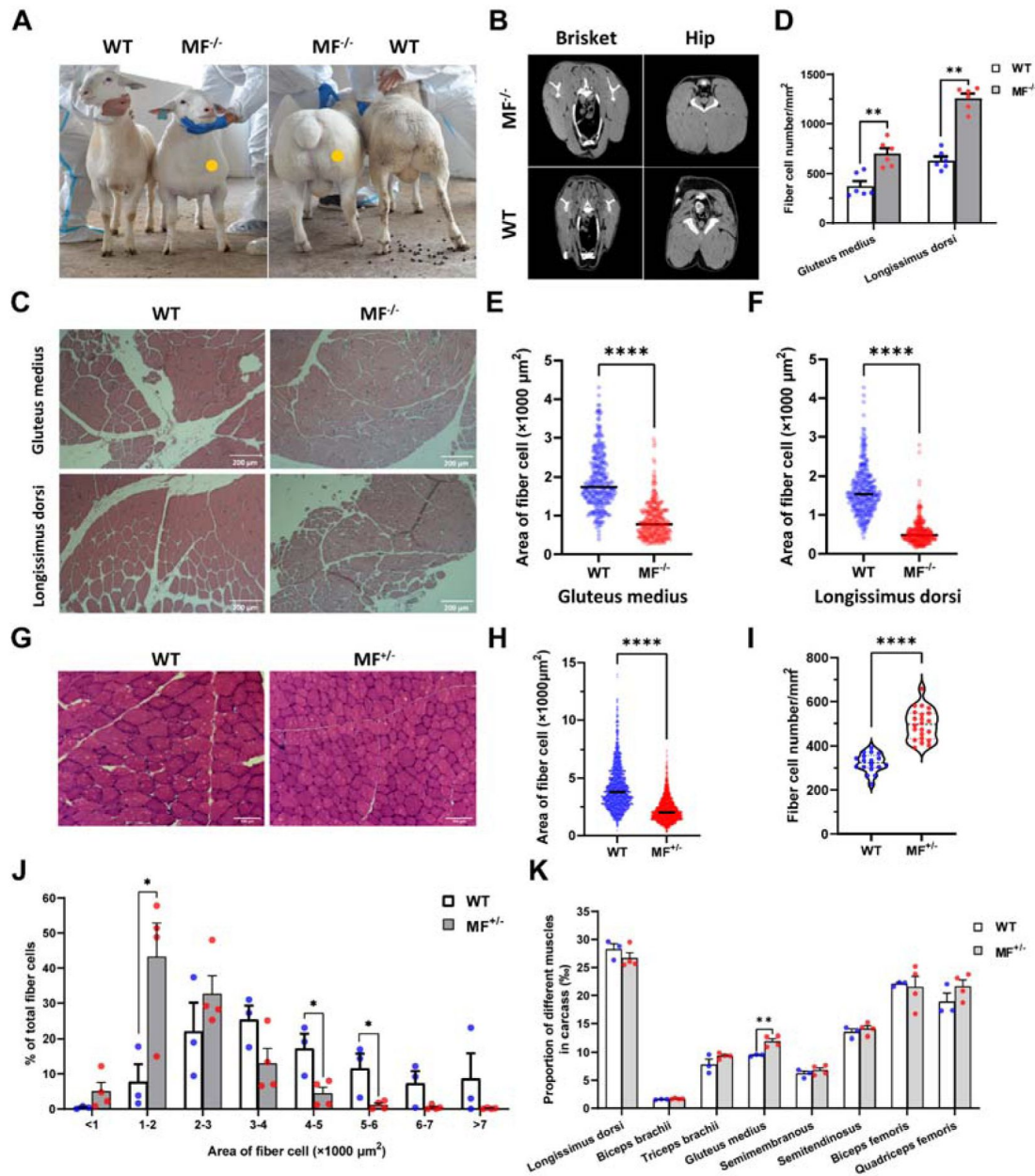


Figure 2

The *MSTN*^{Δel273C} mutation with *FGF5* knockout sheep highlights a dominant “double-muscle” phenotype and muscle fiber hyperplasia

(A) The 6-month-old WT and MF^{-/-} sheep. The genome-edited sheep displayed an obvious “double-muscle” phenotype compared with the WT. (B) The CT scanning image of the brisket and hip of WT and MF^{-/-} sheep. (C) HE sections of gluteus medius and longissimus dorsi of WT and MF^{-/-} sheep. Scale bar 200 μm. (D) Quantification of muscle fibre cell number of per unit area in WT (n=3) and MF^{-/-} (n=1) sheep. All data points were shown. (E-F) Quantification of muscle fibre cell area of gluteus medius and longissimus dorsi in WT (n=3) and MF^{-/-} (n=1) sheep. All data points were shown. (G) HE sections of gluteus medius in WT and MF^{-/-} sheep. Scale bar 100 μm. (H) Quantification of muscle fibre cell area of gluteus medius in WT (n=3) and MF^{-/-} (n=4) sheep. (I) Quantification of muscle fibre cell number of per unit area in WT (n=3) and MF^{-/-} (n=4) sheep. (J) The percentage of cross-sectional area of different size muscle fibers. (K) The proportion of different muscles in carcass in WT (n=3) and MF^{-/-} (n=4) sheep. Data: mean ± SEM. Unpaired student’s t-test were used for statistical analysis after the equal variance test, otherwise the t-test with Welch’s correction were performed. **P* < 0.05, ***P* < 0.01, ****P* < 0.001, and *****P* < 0.0001.

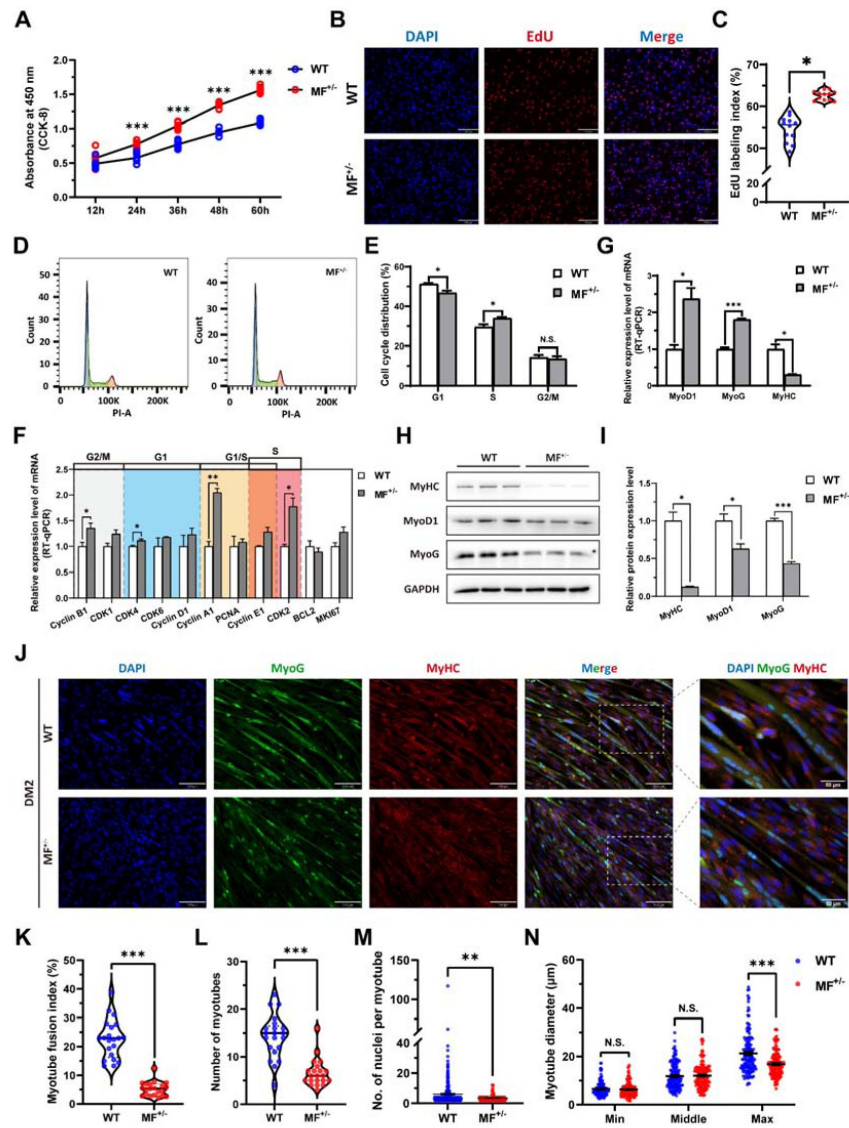


Figure 3

The *MSTN*^{Del273C} mutation with *FGF5* knockout promotes proliferation and inhibits differentiation of skeletal muscle satellite cells

(A) The number of cells was detected by CCK-8 at 12h, 24h, 36h, 48h, and 60h in GM (n=7-8 per group). (B-C) EdU assay showed that the number of EdU positive cells and EdU labeling index were significantly increased in MF^{-/-} cells (n=3). Scale bar 130 μ m. All data points were shown. (D-E) PI staining to detect cell cycle and showed a significant reduce in the proportion of G1 phase and a significant increase in the proportion of S phase in MF^{-/-} cells (n=4). (F) The mRNA expression levels of cell cycle marker genes and cell proliferation marker genes (n=3). (G) The mRNA expression levels of myogenic differentiation marker genes MyoG, MyoD1, and MyHC (n=3). (H-I) The protein expression levels of myogenic differentiation marker genes MyoG, MyoD1, and MyHC (n=3). (J) The MyoG and MyHC immunofluorescence staining of myotubes in DM2. Scale bar 130 μ m. (K) The myotube fusion index, which was represented by the number of cell nuclei in myotubes/total cell nuclei (n=3). All data points were shown. (L) The number of myotubes, which was the number of all myotubes in the field of view (n=3). All data points were shown. (M) The number of nuclei per myotube (n=3). All data points were shown. (N) The myotube diameter (n=3). To reflect the myotube diameter as accurately as possible, the vertical line at the thinnest position of the myotube is taken as the minimum measured (Min), the mid-perpendicular line of the long myotube axis is taken as the middle measured (Middle), and the vertical line at the widest position of the myotube is taken as the maximum measured (Max). All data points were shown. Data: mean $\pm$ SEM. Unpaired student's t-test and chi square test were used for statistical analysis. All student's t-test were performed after the equal variance test, otherwise the t-test with Welch's correction were used. * $P < 0.05$, *** $P < 0.01$, and **** $P < 0.001$.

development; and significantly enriched in MAPK signaling pathway (**Figure 4A-B** [↗](#)). Among DEGs, FOSL1 has aroused our interest. Compared to WT sheep, the FOSL1 mRNA level was significantly ($P<0.05$) lower in both gluteus medius and longissimus dorsi in MF^{+/-} sheep (**Figure 4C** [↗](#)), but it was significantly ($P<0.05$) elevated in MF^{+/-} cells at GM and DM2 (**Figure 4D** [↗](#)). More strikingly, FOSL1 mRNA level was strongly ($P<0.01$) decreased after induced differentiation (**Figure 5E** [↗](#)), and its expression diminished continuously with the differentiation progress. These results suggested that FOSL1 may play a crucial role in the proliferation and myogenic differentiation of skeletal muscle satellite cells.

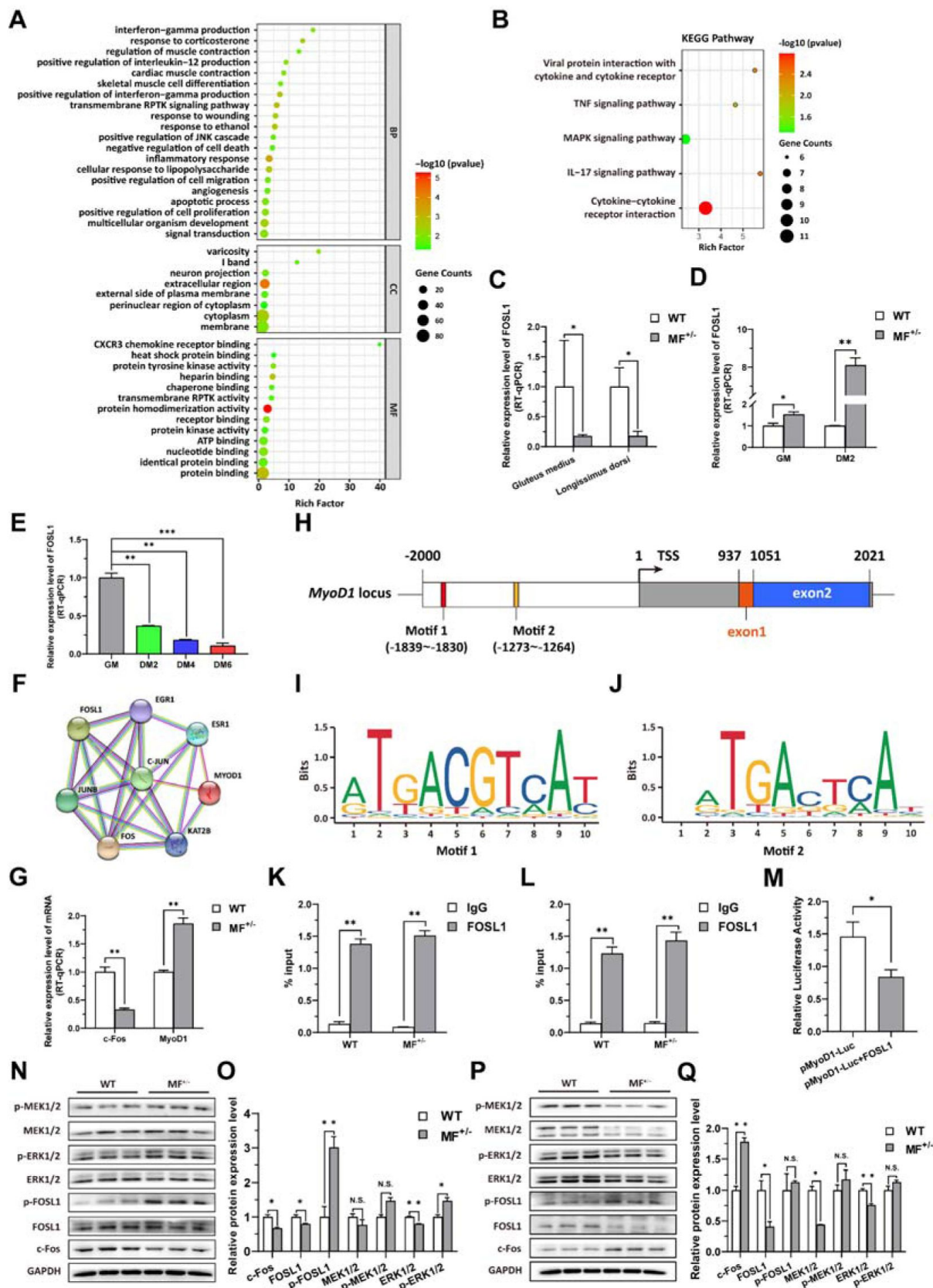


Figure 4

The *MSTN*^{Del273C} mutation with *FGF5* knockout contributes to muscle phenotype via MEK-ERK-FOSL1 axis

(A) Go enrichment analysis of DEGs. Among them, the top 20 entries with significant enrichment are listed in biological process (BP). CC, cellular component; MF, molecular function. (B) KEGG enrichment analysis of DEGs. (C) The mRNA expression level of FOSL1 both at gluteus medius and longissimus dorsi in WT (n=3) and MF^{+/-} (n=4) sheep. (D) The mRNA expression level of FOSL1 both at GM and DM2 in WT and MF^{+/-} cells (n=3). (E) The expression level of FOSL1 mRNA during myogenic differentiation (n=3). (F) The protein-protein interaction (PPI) analysis of FOSL1, c-Fos and MyoD1. (G) The mRNA expression level of c-Fos and MyoD1 at GM in WT and MF^{+/-} myoblasts (n=3). (H) Schematic diagram of MyoD1 gene body, promoter region and binding sites. (I-J) FOSL1 recognition motif in the MyoD1 promoter region. (K) FOSL1 ChIP-qPCR of motif 1 recognition region (n=3). (L) FOSL1 ChIP-qPCR of motif 2 recognition region (n=3). (M) Dual luciferase assay for the effect of FOSL1 on MyoD1 promoter activity (n=4). (N) Western blot of FOSL1, c-Fos, and key kinases of MAPK signaling pathways at GM. (O) Quantification of protein expression of FOSL1, c-Fos, and key kinases of MAPK signaling pathways at GM (n=3). (P) Western blot of FOSL1, c-Fos, and key kinases of MAPK signaling pathways at DM2. (Q) Quantification of protein expression of FOSL1, c-Fos, and key kinases of MAPK signaling pathways at DM2 (n=3). Data: mean ± SEM. Unpaired student's t-test was used for statistical analysis. All student's t-test were performed after the equal variance test, otherwise the t-test with Welch's correction were used. **P* < 0.05, ***P* < 0.01, and ****P* < 0.001.

FOSL1 is a member of the AP-1 family, and another member of this family, c-Fos, inhibits myogenesis and MyoD1 expression by directly binding to the MyoD1 promoter region. Therefore, we speculated that FOSL1 might have similar functions to c-Fos. Subsequent protein-protein interaction (PPI) analysis further suggested that there was a potential interaction between FOSL1 and MyoD1 (Figure 5F). In addition, the c-Fos mRNA level was highly significantly (*P* < 0.01) reduced in MF^{+/-} myoblasts compared to WT cells, whereas the MyoD1 mRNA level was dramatically (*P* < 0.01) increased (Figure 5G). We further found that two bZIP recognition sites in the MyoD1 promoter region had the most significant binding potential to FOSL1 (Figure 5H-J). Subsequently, ChIP-qPCR confirmed that FOSL1 directly binds to these two bZIP recognition sites in the MyoD1 promoter region (Figure 5K-L). The dual luciferase report experiment confirmed that transcription factor FOSL1 can significantly (*P* < 0.05) inhibit MyoD1 promoter activity (Figure 4M). These results indicated that FOSL1 plays an important role in the transcriptional regulation of MyoD1.

As mentioned above, FOSL1 may be involved in the proliferation and myogenic differentiation of skeletal muscle satellite cells. Here, the protein levels of FOSL1 and c-Fos were significantly (*P* < 0.05) reduced in MF^{+/-} cells at GM compared to WT cells (Figure 4N-O), and accompanied by a significant (*P* < 0.05) increase in p-FOSL1 protein levels (Figure 4N-O), whereas FOSL1 protein levels were significantly (*P* < 0.05) diminished and c-Fos protein levels were highly significantly (*P* < 0.01) elevated in MF^{+/-} cells after induced differentiation (Figure 4P-Q), these results further support the key role of FOSL1 on myogenesis. As demonstrated previously, enrichment analysis significantly enriched the MAPK signaling pathway. Compared to WT cells, ERK1/2 protein level was extremely significantly (*P* < 0.01) decreased, and accompanied by a significant (*P* < 0.05) increase in p-ERK1/2 protein levels (Figure 4N-O). After induced differentiation, although both MEK1/2 and ERK1/2 protein levels were dramatically (*P* < 0.01) inhibited (Figure 4P-Q).

Considering the possible serum regulation of MSTN, we examined the effects of MSTN mutations on its receptors and downstream target genes, and observed that both MSTN receptors were significantly up-regulated (Figure S4A), whereas the expression of downstream Smad and Jun families was also inhibited to a varying degree (Figure S4B-C). Furthermore, serum from MF^{+/-} sheep promoted the proliferation of skeletal muscle satellite cells (Figure S4D). *MSTN*^{Del273C}

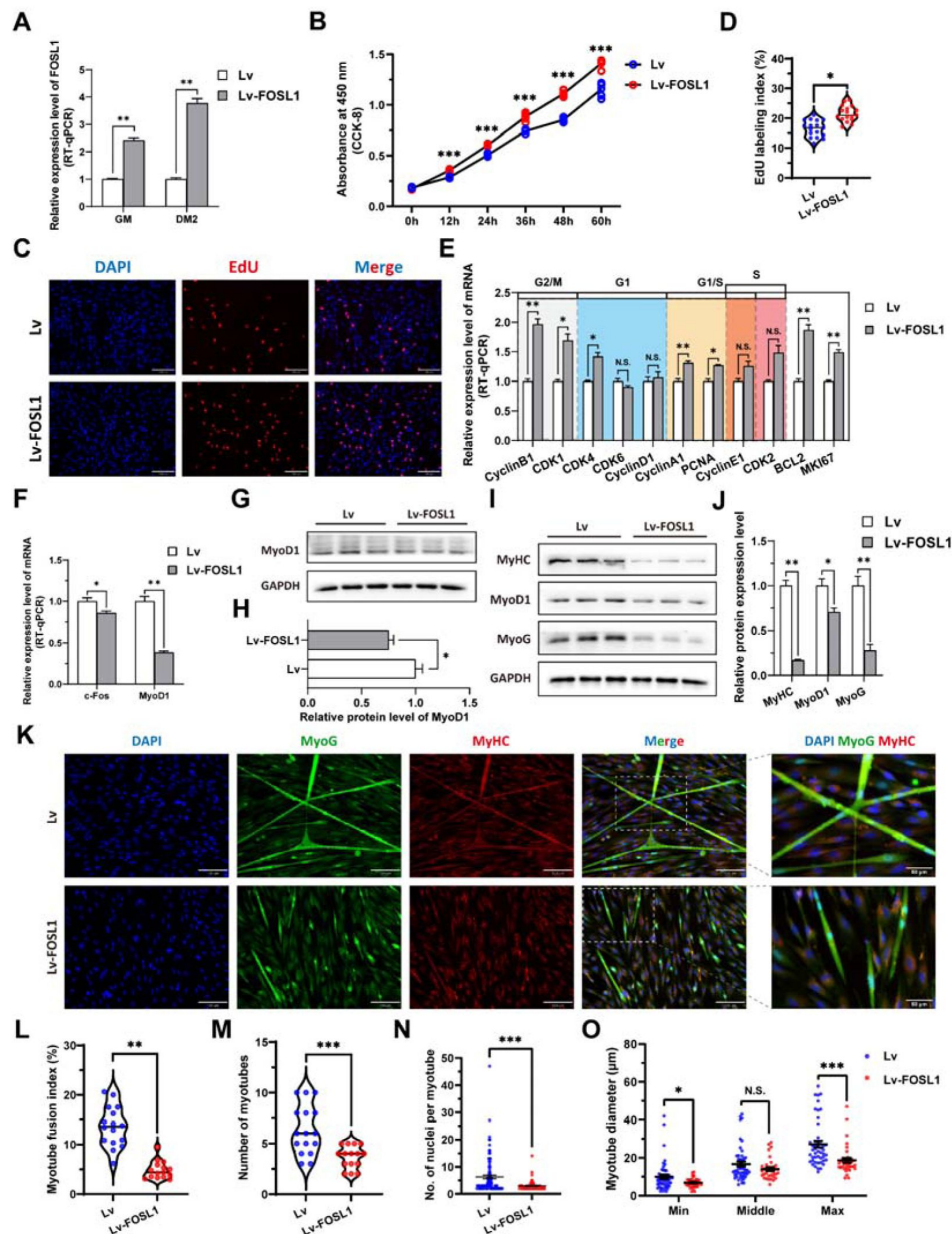


Figure 5

The overexpression of *FOSL1* promotes proliferation and inhibits differentiation of skeletal muscle satellite cells

(A) The mRNA expression level of FOSL1 at GM and DM2 after lentivirus infection (n=3). (B) The number of cells detected by CCK-8 at 0h, 12h, 24h, 36h, 48h, and 60h after infection with lentivirus (n=4-6). (C-D) EdU assay showed that the number of EdU positive cells and EdU labeling index were significantly increased after infection with lentivirus (n=3). Scale bar 130 μ m. All data points were shown. (E) The mRNA expression levels of cell cycle marker genes and cell proliferation marker genes (n=3). (F) The mRNA expression levels of c-Fos and MyoD1 at GM after overexpression of FOSL1 (n=3). (G-H) The protein expression levels of MyoD1 at GM after overexpression of FOSL1 (n=3). (I-J) The protein expression levels of myogenic differentiation marker genes MyoD1, MyoG and MyHC at DM2 after overexpression of FOSL1 (n=3). (K) The MyoG and MyHC immunofluorescence staining of myotubes at DM2 after overexpression of FOSL1. Scale bar 130 μ m. (L-O) The myotube fusion index, number of myotubes, number of nuclei per myotube and the myotube diameter at DM2 after overexpression of FOSL1 (n=3). All data points were shown. Data: mean $\pm$ SEM. Unpaired student's t-test and chi square test were used for statistical analysis. All student's t-test were performed after the equal variance test, otherwise the t-test with Welch's correction were used. * $P < 0.05$, ** $P < 0.01$, and *** $P < 0.001$.

mutation with *FGF5* knockout promoted FOSL1 expression using WT sheep serum (**Figure S4E**), which was similar to the results of FBS culture and HS induction. The serum from MF^{+/-} sheep strongly stimulated FOSL1 expression and the inhibition of MyoD1 (**Figure S4F**). These results suggest that serum regulation cannot be ignored after *MSTN*^{Del273C} mutation with *FGF5* knockout.

In summary, *MSTN*^{Del273C} mutation with *FGF5* knockout may regulate the expression and activity of FOSL1 via the MEK1/2-ERK1/2-FOSL1 axis to affect the proliferation and myogenic differentiation of skeletal muscle satellite cells, and further contribute to muscle phenotype.

3.5 FOSL1 expression and activity control the proliferation and myogenic differentiation of skeletal muscle satellite cells

To investigate the role of FOSL1 on the proliferation and myogenic differentiation of skeletal muscle satellite cells, we successfully constructed *FOSL1* gain-of-function model (**Figure 5A**). We found that overexpression of *FOSL1* significantly ($P<0.05$) promotes cell proliferation (**Figure 5B-E**), suppressed c-Fos mRNA level (**Figure 5F**) ($P<0.05$), and inhibited MyoD1 mRNA ($P<0.01$) and protein ($P<0.05$) levels (**Figure 5F-H**). These results are consistent with what we observed in MF^{+/-} cells at GM, suggesting a potential inhibitory effect of FOSL1 protein on MyoD1. In addition, the protein levels of MyoD1, MyoG and MyHC were all significantly decreased ($P<0.05$), proving that the myogenic differentiation was inhibited after FOSL1 was overexpressed (**Figure 5I-J**). Subsequently, immunofluorescence staining further confirmed the significant ($P<0.05$) inhibitory effect on cell differentiation (**Figure 5K-L**). Also, the number of myotubes, the number of nuclei per myotube, and the myotube diameter were all significantly decreased ($P<0.05$) (**Figure 5K, M-O**). Complementarily, we also constructed *FOSL1* loss-of-function model (**Figure 6A**). In contrast to what observed in FOSL1 overexpression, interference with FOSL1 inhibited skeletal muscle satellite cell proliferation (**Figure 6B-E**), elevated c-Fos and MyoD1 mRNA levels (**Figure 6F**) and significantly ($P<0.05$) increased MyoD1 protein level (**Figure 6G-H**). With myogenic differentiation markers were significantly up-regulated (**Figure 6I-J**), the myotube fusion index, the number of myotubes, the number of nuclei per myotube, and the myotube diameter were all significantly increased (**Figure 6K-O**). These results further demonstrated that elevated FOSL1 level inhibits myogenic differentiation and produces smaller myotubes.

To further ascertain this insight, the tert-butylhydroquinone (TBHQ), which can strongly activate ERK1/2 and increase p-ERK1/2 protein expression level, was used to activate ERK1/2 and act as an indirect activator of FOSL1. As expected, the addition of 20 μ M TBHQ significantly ($P<0.01$) inhibited the myogenic differentiation of skeletal muscle satellite cells (**Figure 7A-B**). And, the number of myotubes was significantly increased ($P<0.05$) (**Figure 7A, C**), while the number of nuclei per myotube was significantly ($P<0.05$) decreased and produced a smaller myotube diameter ($P<0.05$) (**Figure 7A, D-E**). In addition, PD98509, which can inhibit ERK1/2 and dose-dependent reduce p-ERK1/2 protein expression level, was used for complementary experiments. We observed opposite results to TBHQ treatment, including increase in myotube fusion index, the number of nuclei per myotube, and the myotube diameter (**Figure 7F-J**). Taken together, our results shed light that the *MSTN*^{Del273C} mutation with *FGF5* knockout mediated the activation of FOSL1 via MEK-ERK-FOSL1 axis, further promotes skeletal muscle satellite cell proliferation, and inhibits myogenic differentiation by inhibiting the transcription of MyoD1, and resulting in smaller myotubes.

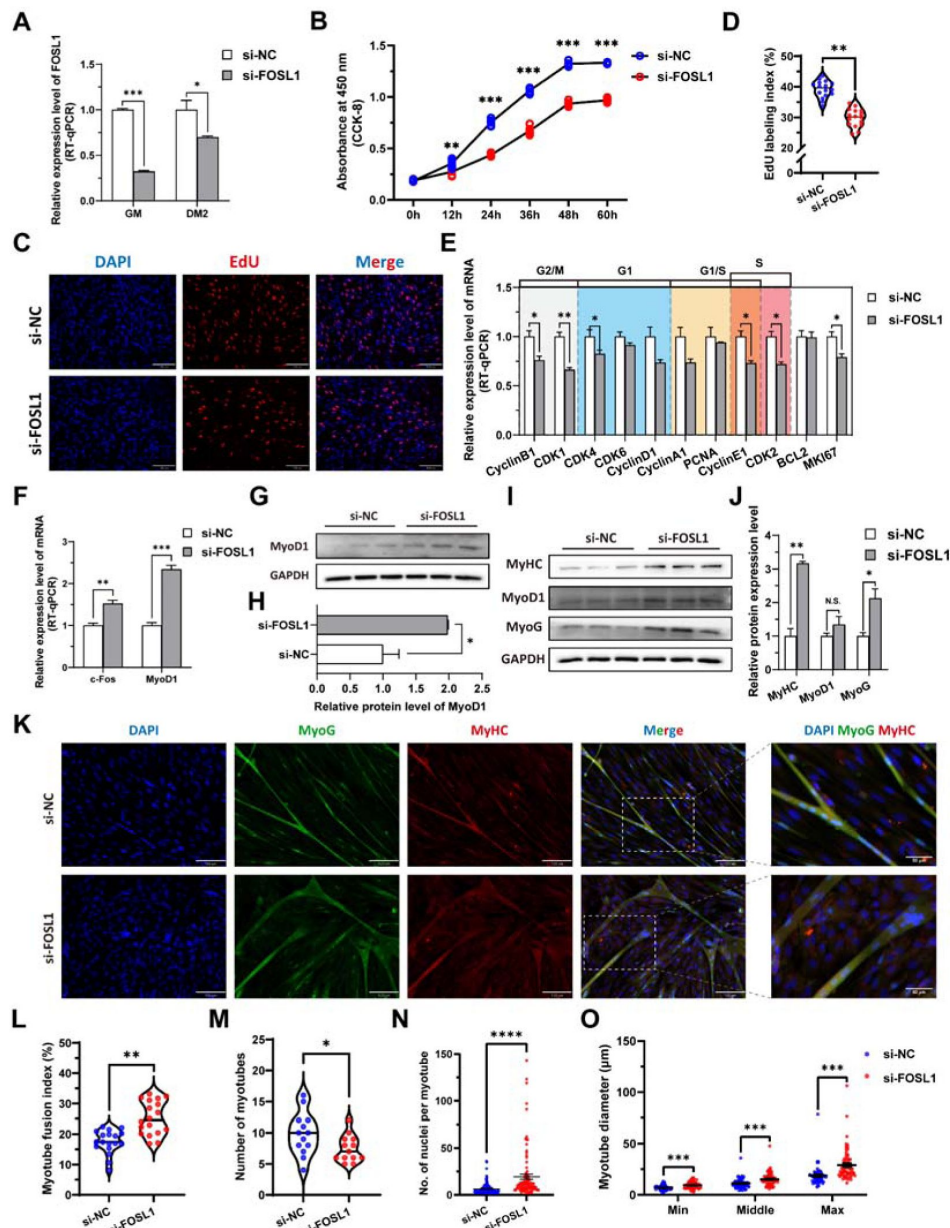


Figure 6

The inhibition of *FOSL1* suppresses proliferation and promotes differentiation of skeletal muscle satellite cells

(A) The mRNA expression level of *FOSL1* at GM and DM2 after inhibiting *FOSL1* ($n=3$). (B) The number of cells detected by CCK-8 at 0h, 12h, 24h, 36h, 48h, and 60h after inhibiting *FOSL1* ($n=4-6$). (C-D) EdU assay showed that the number of EdU positive cells and EdU labeling index were significantly decreased after inhibiting *FOSL1* ($n=3$). Scale bar 130 μ m. All data points were shown. (E) The mRNA expression levels of cell cycle marker genes and cell proliferation marker genes ($n=3$). (F) The mRNA expression levels of *c-Fos* and *MyoD1* at GM after inhibiting *FOSL1* ($n=3$). (G-H) The protein expression levels of *MyoD1* at GM after inhibiting *FOSL1* ($n=3$). (I-J) The protein expression levels of myogenic differentiation marker genes *MyoD1*, *MyoG* and *MyHC* at DM2 after inhibiting *FOSL1* ($n=3$). (K) The *MyoG* and *MyHC* immunofluorescence staining of myotubes at DM2 after inhibiting *FOSL1*. Scale bar 130 μ m. (L-O) The myotube fusion index, number of myotubes, number of nuclei per myotube and the myotube diameter at DM2 after overexpression of *FOSL1* ($n=3$). All data points were shown. Data: mean $\pm$ SEM. Unpaired student's t-test and chi square test were used for statistical analysis. All student's t-test were performed after the equal variance test, otherwise the t-test with Welch's correction were used. * $P < 0.05$, ** $P < 0.01$, *** $P < 0.001$, and **** $P < 0.0001$.

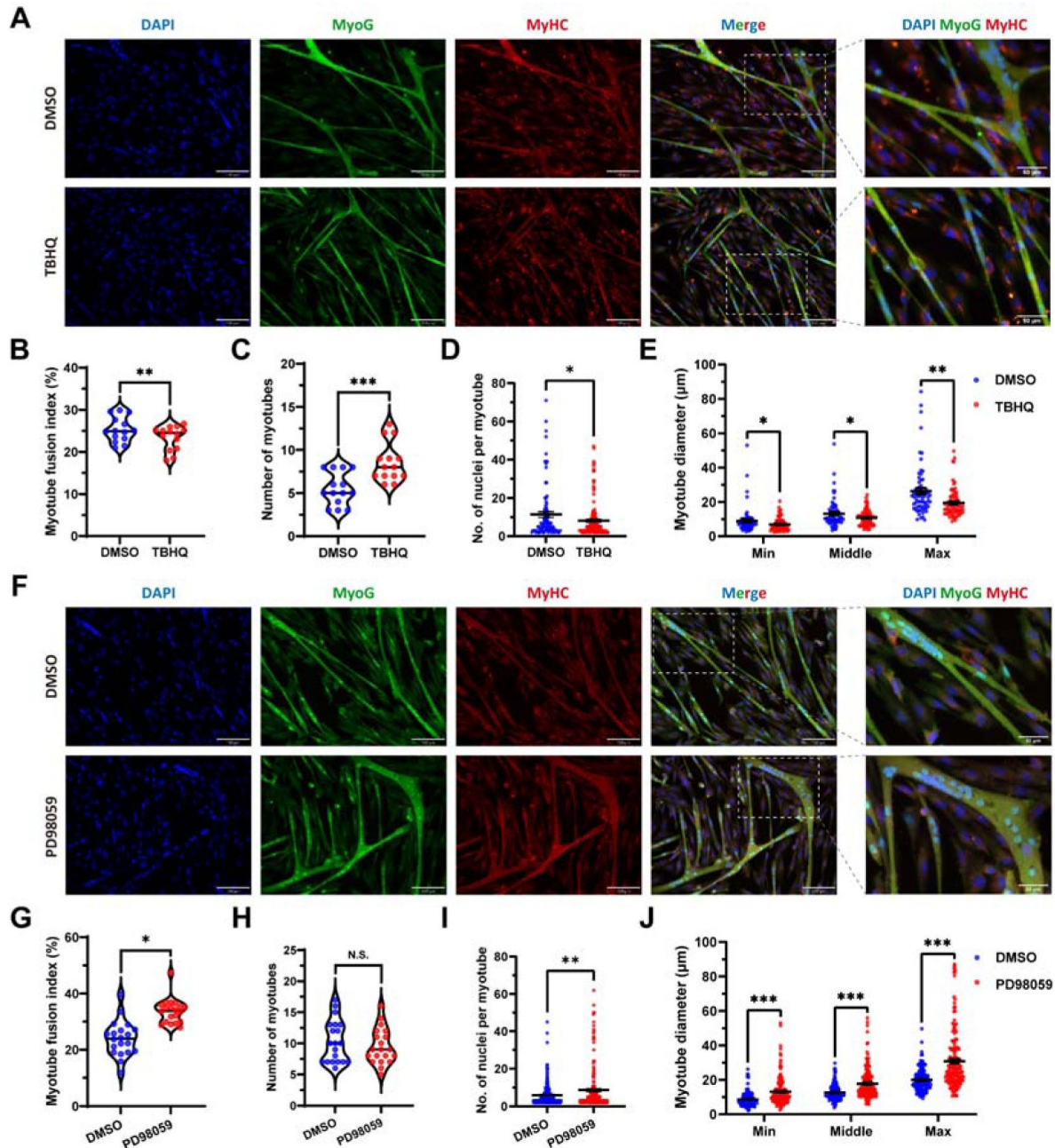


Figure 7

FOSL1 activity is a key regulator of myogenic differentiation and muscle myofiber hyperplasia

(A) Immunofluorescence staining of myogenic differentiation markers MyoG and MyHC in sheep skeletal muscle satellite cells at DM2 after addition of 20 μ M TBHQ. Scale bar 130 μ m or 50 μ m. (B-E) The myotube fusion index, number of myotubes, number of nuclei per myotube and the myotube diameter at DM2 after addition of 20 μ M TBHQ (n=3). All data points were shown. (F) Immunofluorescence staining of myogenic differentiation markers MyoG and MyHC in sheep skeletal muscle satellite cells at DM2 after addition of 1 μ M PD98059. Scale bar 130 μ m or 50 μ m. (G-J) The myotube fusion index, number of myotubes, number of nuclei per myotube and the myotube diameter at DM2 after addition of 1 μ M PD98059 (n=3). All data points were shown. Data: mean $\pm$ SEM. Unpaired student's t-test and chi square test were used for statistical analysis. All student's t-test were performed after the equal variance test, otherwise the t-test with Welch's correction were used. * P < 0.05, ** P < 0.01, and *** P < 0.001.

3.6 The *MSTN*^{Del273C} mutation with *FGF5* knockout inhibit calcium-dependent transcription signal pathway to regulate secondary fusion of myotubes

As mentioned previously, DEGs identified by RNA-seq were significantly enriched in biological processes such as muscle contraction and cardiac muscle contraction. In muscle contraction, calcium ions (Ca^{2+}) play a crucial role, triggering the process of muscle contraction and relaxation by binding to proteins in muscle fibers. Here, the calcium-calmodulin dependent protein kinase II (CaMKII) α/δ protein level was significantly ($P < 0.05$) reduced in $\text{MF}^{+/-}$ cells compared to WT cells (Figure 8A-B), and the intracellular Ca^{2+} concentration was also significantly ($P < 0.05$) reduced (Figure 8C-D). The high-throughput Ca^{2+} channel RYR is responsible for rapid and massive release Ca^{2+} from the endoplasmic reticulum into cytoplasm. We observed a significant ($P < 0.05$) decrease in RYR1 and/or RYR3 mRNA levels in $\text{MF}^{+/-}$ sheep skeletal muscle satellite cells and myotubes cells (Figure 8E-F). In addition, intracellular Ca^{2+} concentration correlated with myoblasts fusion, whereas the mRNA levels of MYMK and MYMX, which control myoblasts fusion, were significantly ($P < 0.01$) reduced in $\text{MF}^{+/-}$ cells (Figure 8E-F). These results suggest that the decrease Ca^{2+} levels and inhibition of myoblasts fusion genes may be potential triggers for the decrease of myotube diameter and myofiber cross-sectional area in $\text{MF}^{+/-}$ sheep.

In a word, our results shed light that the *MSTN*^{Del273C} mutation with *FGF5* knockout mediated the activation of FOSL1 via MEK-ERK-FOSL1 axis (Figure 9). The activated FOSL1 promotes skeletal muscle satellite cell proliferation and inhibits myogenic differentiation by inhibiting the expression of MyoD1, and resulting in fusion to form smaller myotubes (Figure 9). In addition, activated ERK1/2 may inhibit the secondary fusion of myotubes by Ca^{2+} -dependent CaMKII activation pathway, leading to myoblasts fusion to form smaller myotubes (Figure 9).

4 Discussion

4.1 Optimized Cas9 mRNA and sgRNA delivery ratio improves the efficiency of dual-gene biallelic homozygous mutations

The strategy for producing gene knockout animals by CRISPR/Cas9 gene editing system is usually to introduce the Cas9 mRNA and the sgRNA of the target gene into their prokaryotic embryos by microinjection. However, this “one-step” method often results in a “mosaic” of gene-edited offspring (Wan *et al.*, 2015). Such chimeric mutants have now been reported in gene knockout mice (Wang *et al.*, 2013), rats (Bao *et al.*, 2015), monkeys (Niu *et al.*, 2014), pigs (Hai *et al.*, 2014), sheep (Hongbing HAN, 2014), goats (Wang *et al.*, 2015), rabbits (Lv *et al.*, 2016), and humans (Wang and Yang, 2019) prepared by a “one-step” method using the CRISPR/Cas9 system. For studies involved in genetic phenotypes, chimeric gene knockout animals require further cross-breeding to obtain animals with a complete knockout of the target gene. Once required to generate multiple gene knockout animals, this time-consuming and laborious operation will become extremely difficult. Although many studies have been devoted to eliminating this widespread chimeric mutation (Sato *et al.*, 2015; Sung *et al.*, 2014; Kotani *et al.*, 2015; Chen *et al.*, 2015; Zhou *et al.*, 2014; Tu *et al.*, 2017; Wang *et al.*, 2015), however, these optimizations did not bring about a significant improvement in the production efficiency of biallelic knockout animals. Here, we increased the delivery ratio of Cas9 mRNA to sgRNA from 1:2 to 1:10, which improve the efficiency of the homozygous mutation of the biallelic gene. This unprecedented optimization method not only improved the overall gene knockout efficiency, but also the obtained gene-edited offspring were all dual-gene biallelic mutation. However, it is necessary to point out that although there are statistical differences, due to the limited number of sheep we actually

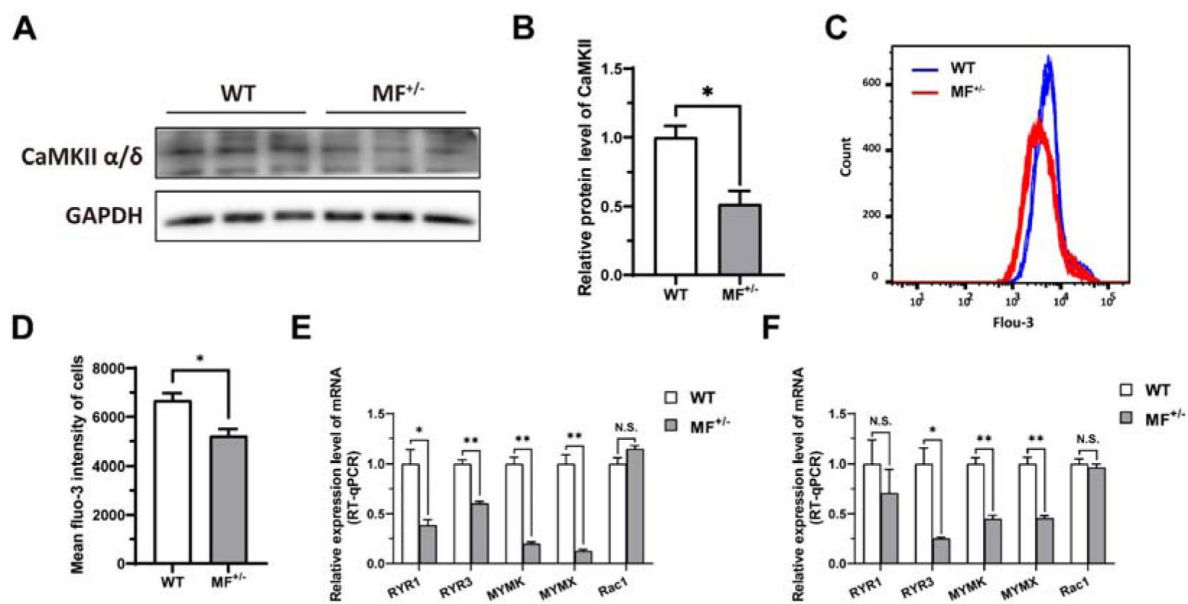


Figure 8

The *MSTN*^{Del273C} mutation with *FGF5* knockout inhibit calcium-dependent transcription signal pathway

(A-B) The protein expression level of CaMKII α/δ between WT and MF^{+/-} cells at GM (n=3). (C) Distribution of intracellular Ca²⁺ signals between WT and MF^{+/-} cells at GM. (D) Average intracellular Ca²⁺ fluorescence intensity between WT and MF^{+/-} cells at GM (n=4). (E) The mRNA expression levels of Ca²⁺ channels and myoblast fusion-related genes at GM (n=3). (F) The mRNA expression levels of Ca²⁺ channels and myoblast fusion-related genes at DM2 (n=3). Data: mean $\pm$ SEM. Unpaired student's t-test was used for statistical analysis. All student's t-test were performed after the equal variance test, otherwise the t-test with Welch's correction were used. * $P < 0.05$, ** $P < 0.01$.

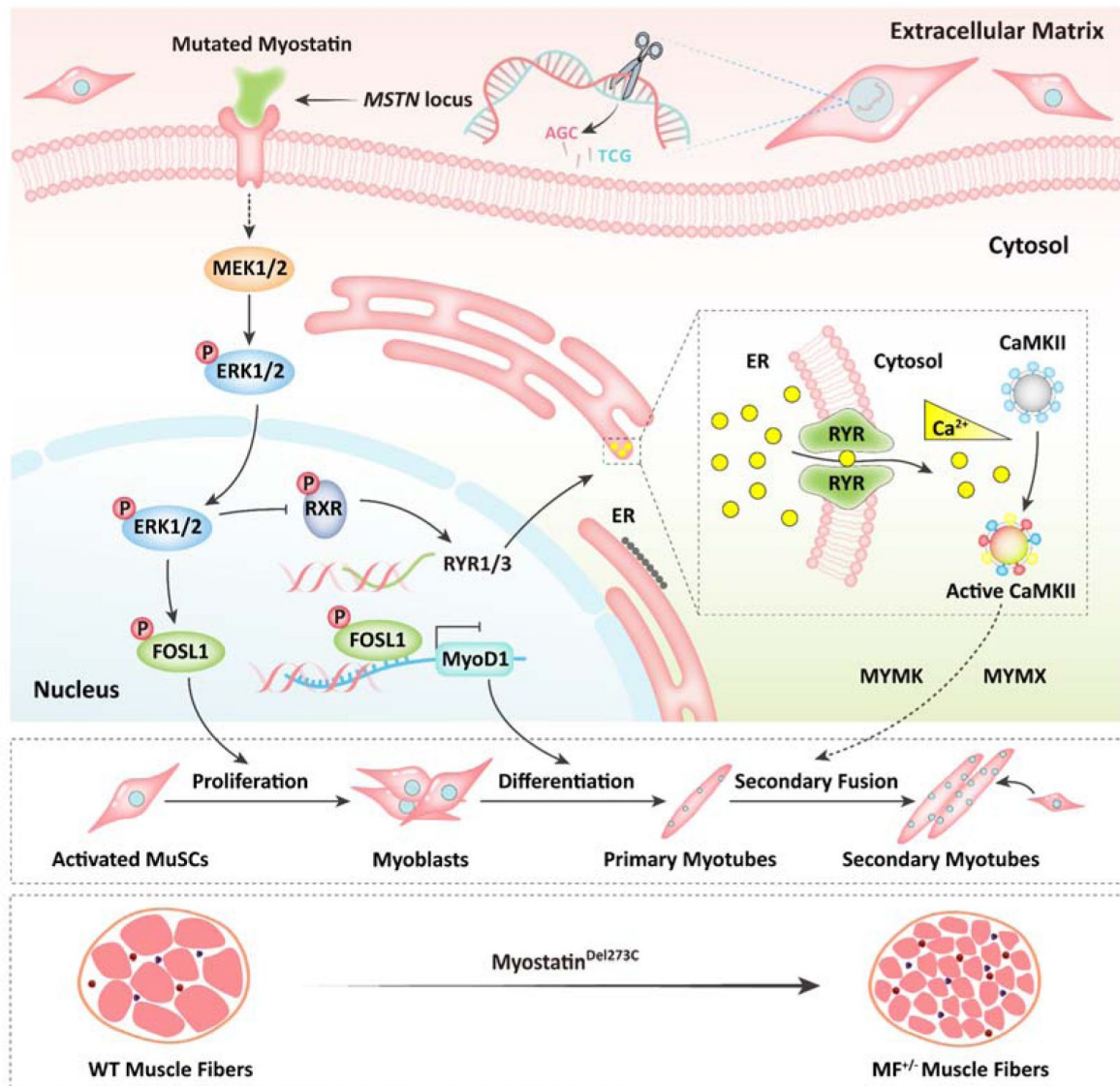


Figure 9

Schematic illustration of the regulation of muscle phenotypes by *MSTN*^{Del273C} mutation with *FGF5* knockout

The *MSTN*^{Del273C} mutation with *FGF5* knockout mediated the activation of FOSL1 via MEK-ERK-FOSL1 axis. The activated FOSL1 promotes skeletal muscle satellite cell proliferation and inhibits myogenic differentiation by inhibiting the expression of MyoD1, and resulting in fusion to form smaller myotubes. In addition, activated ERK1/2 may inhibit the secondary fusion of myotubes by Ca²⁺-dependent CaMKII activation pathway, leading to myoblasts fusion to form smaller myotubes.

produced and used for evaluation, the strategy to improve the efficiency of the homozygous mutation of biallelic gene by increasing the Cas9 and mRNA delivery ratio needs to be further confirmed in future studies.

4.2 Phenotypes produced by *MSTN* mutations are mutation site-dependent

As mentioned previously, although *MSTN* mutations have been found to produce a “double-muscle” phenotype in multiple species, the microscopic phenotypes are different, and this difference is closely related to the mutation site and species types. In mice, the number of skeletal muscle fibers with *MSTN* gene knockout significantly increased by 86% (McPherron *et al.*, 1997 [\[1\]](#)). A missense mutant *MSTN* only increased the number of mouse muscle fibers, while dominant negative *MSTN* resulted in increased muscle fiber cross-sectional area in mice, but not the number of muscle fibers (Nishi *et al.*, 2002 [\[2\]](#); Zhu *et al.*, 2000 [\[3\]](#)). In addition, the use of *MSTN* neutralizing antibody on adult rats also resulted in an increased muscle fiber cross-sectional area (Haidet *et al.*, 2008 [\[4\]](#)). In cattle, natural *MSTN* mutant Belgian Blue cattle had an increased number of muscle fibers and reduced muscle fiber diameter (Wegner *et al.*, 2000 [\[5\]](#)). The muscle fiber cross-sectional area of longissimus dorsi and gluteus medius in sheep was significantly increased after a 4bp deletion of the first exon of *MSTN* (Zhiliang *et al.*, 2004 [\[6\]](#)). In pigs, both the *MSTN* gene-edited Meishan and Hubei pigs showed a phenotype with increased muscle fiber density (Qian *et al.*, 2015 [\[7\]](#); Xu *et al.*, 2013 [\[8\]](#)). Here, we prepared *MSTN*^{Del273C} mutation with *FGF5* knockout sheep with 3-base pairs of AGC in the third exon of *MSTN*, which caused the deletion of cysteine at amino acid position 273. Its macroscopic phenotype is similar to that of the *MSTN*-edited sheep with the first exon knocked out 4-base pairs. Both of them showed an abnormally developed “double-muscle” phenotype of hip muscle, but the *MSTN*^{Del273C} mutation with *FGF5* knockout sheep highlights a muscle fiber hyperplasia phenotype.

4.3 FOSL1 recognizes and binds to the MyoD1 promoter and inhibits its expression

As previously described, AP-1 family members play key roles in skeletal muscle cell proliferation, differentiation, and muscle development. In this study, AP-1 family member *FOSL1* was significantly reduced in MF^{+/-} sheep, and its expression was drastically reduced during myogenic differentiation, which was consistent with the decrease of *FOSL1* expression during C2C12 differentiation (Tobin *et al.*, 2016 [\[9\]](#)). In addition, FOSL2, another AP-1 family member, can also inhibit myoblast differentiation (Alli *et al.*, 2013a [\[10\]](#)), which may support the inhibitory effect of FOSL1 on myogenic differentiation. Therefore, *FOSL1* was recognized as a potential gatekeeper. It has been shown that *FOSL1* heterodimerizes with other transcription factors, such as the members of the bZIP family, and these dimers are either disabling the transcriptional activator complex or saving the interacting proteins from degradation in proteasomes (Sobolev *et al.*, 2022 [\[11\]](#)). Moreover, c-Fos, a member of the AP-1 family, has been shown to inhibit MyoD1 expression and myogenesis by directly binding to the MyoD1 promoter region (Li *et al.*, 1992 [\[12\]](#)). Therefore, we speculate that *FOSL1* may have similar functions to c-Fos. PPI analysis suggested a potential interaction between *FOSL1* and MyoD1. Subsequently, we confirmed that FOSL1 directly binds to two bZIP recognition sites in the MyoD1 promoter region, and inhibits MyoD1 promoter activity. Meanwhile, the overexpression and interference of FOSL1 further confirmed the inhibitory effect of FOSL1 on the expression of MyoD1. In a word, these results fully support our hypothesis that FOSL1 binds the MyoD1 promoter and inhibits its expression.

4.4 The *MSTN*^{Del273C} mutation with *FGF5* knockout contribute to muscle phenotype via MEK-ERK-FOSL1 axis

The nonclassical pathway of MSTN involves PI3K/Akt/mTOR signaling pathway and MAPK signaling pathway, which mainly includes ERKs, JNKs and p38 MAPK (Huang *et al.*, 2007 [\[1\]](#); Gui *et al.*, 2012 [\[2\]](#)). All of those pathways are involved in the signal transduction pathway of MSTN and mediate the transcription of MRFs (Myogenin, Myf5, MyoD), MuRF-1 and Atrogin-1, to regulate myogenic differentiation and skeletal muscle quality (Chen *et al.*, 2021b [\[3\]](#)). *MSTN* induces muscle fiber hypertrophy prior to satellite cell activation (Wang and McPherron, 2012 [\[4\]](#)) and inhibits IGF-I-induced increase in myotube diameter through Akt signaling pathway (Morissette *et al.*, 2009 [\[5\]](#)). In our study, the *MSTN*^{Del273C} mutation with *FGF5* knockout resulted in the inhibition of myogenic differentiation of skeletal muscle satellite cells, and the number of myotubes and the myotube size were significantly reduced.

As previously described, the DEGs of gluteus medius RNA-seq were significantly enriched in the MAPK signaling pathway. In fact, the MAPK signaling pathway has been proven to be closely related to muscle development and myoblast differentiation (Xie *et al.*, 2018 [\[6\]](#); Segales *et al.*, 2016 [\[7\]](#)). For example, ERK1/2 promotes myoblast proliferation in response to various growth factors (Campbell *et al.*, 1995 [\[8\]](#)), inhibits signaling pathways that activate ERK1/2, or isolates ERK1/2 in the cytoplasm, leading to cell cycle exit and cell differentiation (Jones *et al.*, 2001 [\[9\]](#); Michailovici *et al.*, 2014 [\[10\]](#)). The RXR activity in myoblasts promotes myogenesis by regulating MyoD expression and acting as a MyoG cofactor (Zhu *et al.*, 2009 [\[11\]](#)). Inhibition of MEK1/2 activates satellite cell differentiation in primary muscle fibers (Alli *et al.*, 2013b [\[12\]](#)), and also induces myogenic differentiation and excessive fusion (Eigler *et al.*, 2021 [\[13\]](#)). A recent study on glioma showed that FOSL1 can be activated by the Ras-MEK1/2-ERK1/2 axis in MAPK signaling pathway (Marques *et al.*, 2021 [\[14\]](#)). Similarly, the activated MEK1/2-ERK1/2 axis in aged skeletal muscle also activates FOSL1 and increases the abundance of FOSL1 and the trans-activation capacity of the Fos-Jun heterodimer (Mathes *et al.*, 2021 [\[15\]](#)). In our study, the *MSTN*^{Del273C} mutation with *FGF5* knockout regulates FOSL1 expression and activity through MEK1/2-ERK1/2-FOSL1 axis and activated FOSL1 further inhibits myogenic differentiation of skeletal muscle satellite cells, resulting in smaller myotube diameter. However, despite the high expression of p-FOSL1 in MF^{+/+} myoblasts, it did not significantly inhibit the transcription of MyoD1, which may be related to a dramatic enhance in c-Fos, or there might be other parallel signaling pathways regulating MyoD1 after *MSTN*^{Del273C} mutation with *FGF5* knockout.

Furthermore, it has been demonstrated that the inhibition of MEK1/2 using MEK1/2-specific inhibitor PD184352 can significantly down-regulate FOSL1 expression (Mathes *et al.*, 2021 [\[15\]](#)). In our study, the indirect activation of FOSL1 by TBHQ can inhibit the myogenic differentiation of sheep skeletal muscle satellite cells, leading to reduced myoblast fusion capacity and smaller myotube diameter. In contrast, inhibition of the MEK1/2 pathway by PD98059 to suppress FOSL1 activity produced the opposite effect. Taken together, these results shed light on the potential mechanisms by which *MSTN*^{Del273C} mutation with *FGF5* knockout leads to increased myofiber numbers and decreased fiber cross-sectional area.

4.5 The *MSTN*^{Del273C} mutation with *FGF5* knockout may affect myoblasts fusion through ERK-mediated calcium dependent transcriptional signaling pathway

Ca²⁺ is recognized as a regulator of mammalian muscle fusion (Eigler *et al.*, 2021 [\[13\]](#); Constantin *et al.*, 1996 [\[16\]](#)). The transient depletion of Ca²⁺ in the endoplasmic reticulum is associated with myoblast differentiation and fusion (Jones *et al.*, 2001 [\[9\]](#)). However, the signaling cascade leading to Ca²⁺-mediated myoblast fusion remains unclear. Intracellular Ca²⁺ level is regulated by various

Ca^{2+} voltage-gated channels, including but not limited to ryanodine receptors (RyR), which is the main Ca^{2+} release channel of the sarcoplasmic reticulum. CaMKII is a member of the calcium/calmodulin-dependent serine/threonine kinase family. CaMKII δ , CaMKII γ , and CaMKII β are the main isoforms expressed in skeletal muscle (Bayer *et al.*, 1996 [↗](#)). Recent studies have shown that elevated intracellular Ca^{2+} level is crucial for myoblasts fusion and that Ca^{2+} signaling in newly formed myotubes occurs prior to the rapid growth stage of myotubes, indicating that Ca^{2+} released from the endoplasmic reticulum in early myotubes may promote secondary fusion of myoblasts and myotube expansion (Eigler *et al.*, 2021 [↗](#)). Further studies showed that Ca^{2+} -dependent activation of CaMKII is essential for myotubes expansion and may mediate myoblast-myotube fusion by regulating MYMK and Rac1, but it is not necessary for myoblast-myoblast fusion (Eigler *et al.*, 2021 [↗](#)).

In our study, the *MSTN*^{Del273C} mutation with *FGF5* knockout resulted in a significant reduction of Ca^{2+} level and CaMKII α/δ protein level, and led to a decrease in myotube fusion capacity. Therefore, our results support that the increased p-ERK1/2 level promotes cell proliferation and inhibits myogenic differentiation in MF^{+/−} sheep skeletal muscle satellite cells. Meanwhile, activated ERK1/2 further inhibited RyR activity by suppressing the phosphorylation of RYR, thereby reducing the release of endoplasmic reticulum Ca^{2+} and potentially inhibiting the secondary fusion of myotubes by Ca^{2+} -dependent CaMKII activation pathway, and further mediating myofiber hyperplasia.

5 Conclusion

In this study, we found that increasing the delivery ratio of Cas9 mRNA to sgRNA can improve the efficiency of the homozygous mutation of the biallelic gene. Based on this, we generated a *MSTN*^{Del273C} mutation with *FGF5* knockout sheep, a dual-gene biallelic homozygous mutant, which highlights a dominant “double-muscle” phenotype. Both F0 and F1 generation mutants highlight the excellent trait of high-yield meat and the more number of muscle fibers per unit area. Our results suggested the *MSTN*^{Del273C} mutation with *FGF5* knockout mediated the activation of FOSL1 via MEK-ERK-FOSL1 axis, further promotes skeletal muscle satellite cell proliferation, and inhibits myogenic differentiation by inhibiting the expression of MyoD1, and resulting in smaller myotubes. In addition, activated ERK1/2 may inhibit the secondary fusion of myotubes by Ca^{2+} -dependent CaMKII activation pathway, leading to myoblasts fusion to form smaller myotubes. This supports the myofiber hyperplasia that more number of muscle fibers and smaller cross sectional area, caused by the *MSTN*^{Del273C} mutation with *FGF5* knockout.

Data availability statement

The raw sequence data reported in this paper have been deposited in the Genome Sequence Archive (Genomics, Proteomics & Bioinformatics 2021) in National Genomics Data Center (Nucleic Acids Res 2022), China National Center for Bioinformation / Beijing Institute of Genomics, Chinese Academy of Sciences (GSA: CRA008539) that are publicly accessible at <https://ngdc.cncb.ac.cn/gsa> [↗](#).

Ethics statement

All experiments were performed in accordance with relevant guidelines and adhere to the ARRIVE guidelines (<https://arriveguidelines.org/> [↗](#)) for the reporting of animal experiments. All sheep are raised in accordance with the national feeding standard NT/T815-2004. All procedures performed were consistent with the National Research Council Guide for the Care and Use of Laboratory

Animals. All experimental animal protocols were approved and performed in accordance with the requirements of the Animal Care and Use Committee at China Agricultural University (AW02012202-1-3).

Competing financial interests

The authors declare that there are no competing financial interests.

Author contributions

MMC performed the majority of experiments, data analysis, and drafted the manuscript. YZ performed a part of experiments and revised the manuscript. XLX, SJW and ZML helped with data analysis. XSZ, JLZ and XFG were responsible for the management of the feeding plant, slaughtering, and collecting samples. YMY helped to process some biological information data. SYQ, GY, SQW, HXL and AWW helped to collect and organize original data. GSL led the prokaryotic injection and embryo transfer. YL prepared the gene editing sheep. KY and HBH participated in project management. KY, FHL and ZXL conceived the project, revised manuscript and final approval of manuscript. All authors read and approved the final manuscript.

Acknowledgements

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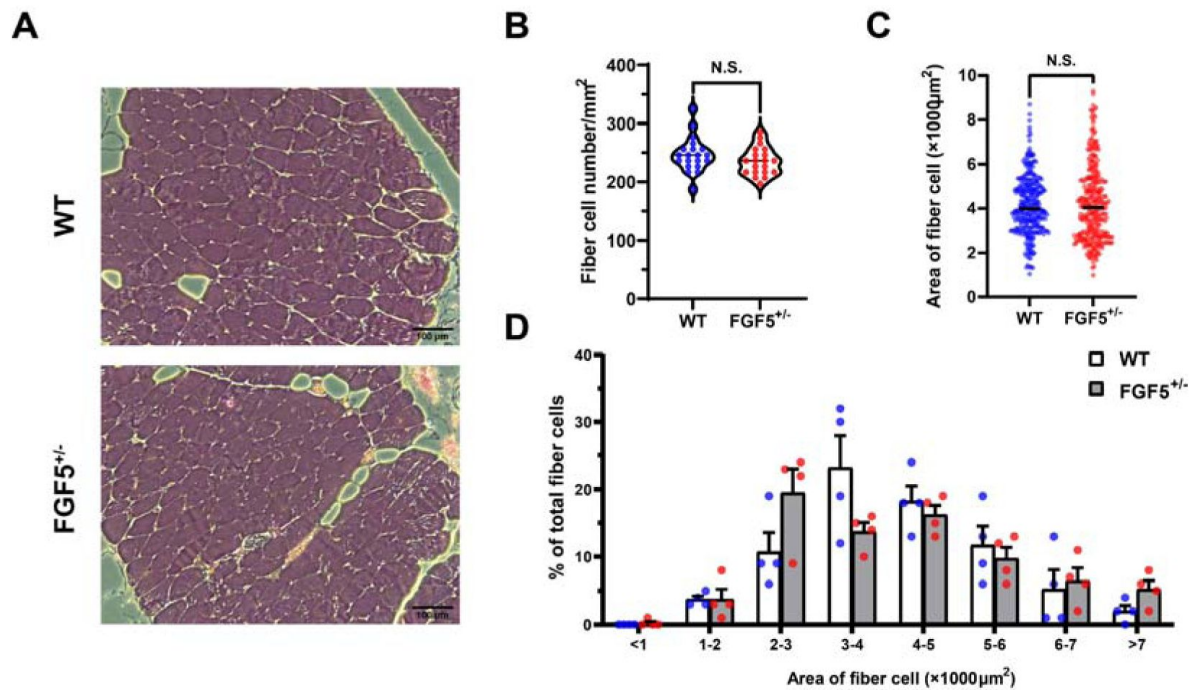


Figure S1

***FGF5* mutation does not affect muscle fiber size**

(A) HE sections of gluteus medius in WT and *FGF5*^{+/-} sheep. Scale bar 100 μm. (B) Quantification of muscle fibre cell area of gluteus medius in WT and *FGF5*^{+/-} sheep.

(C) Quantification of muscle fibre cell number of per unit area in WT and *FGF5*^{+/-} sheep. (D) The percentage of cross-sectional area of different size muscle fibers. Data: mean ± SEM. Unpaired student's t-test was used for statistical analysis. All student's t-test were performed after the equal variance test, otherwise the t-test with Welch's correction were used.

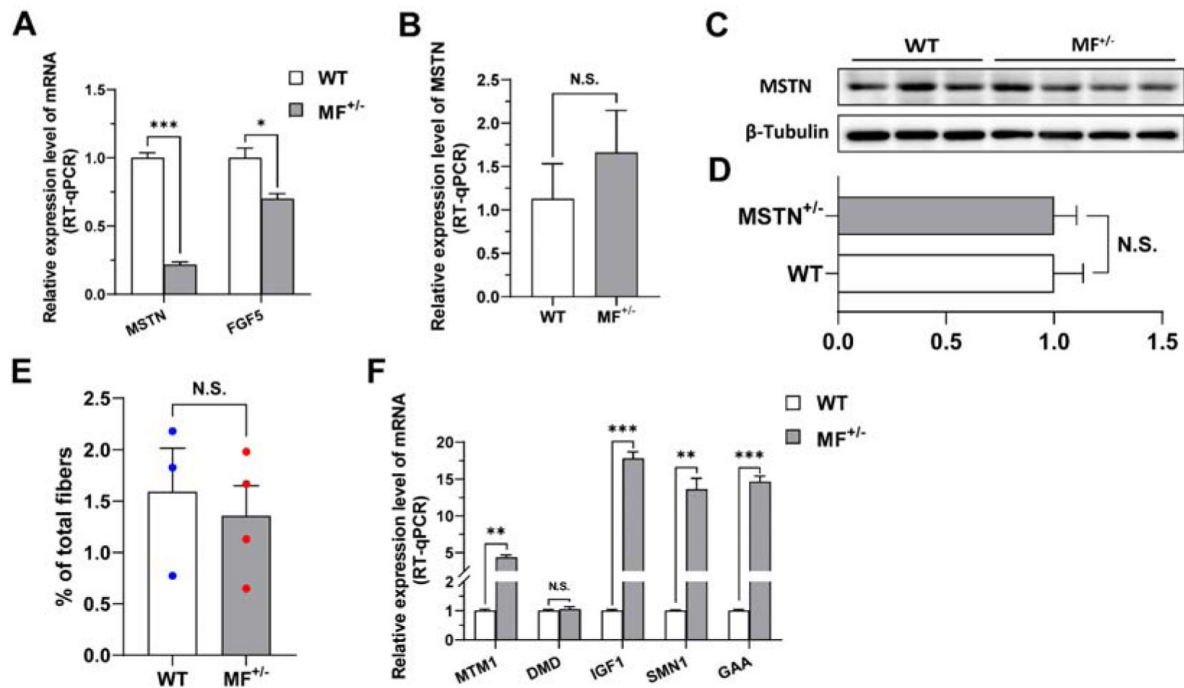


Figure S2

The *MSTN*^{Del273C} mutation with *FGF5* knockout has no potential effect on *MSTN* expression and muscular dystrophy

(A) The mRNA expression levels of *MSTN* and *FGF5* of gluteus medius in WT (n=4) and *MF*^{+/-} (n=4) sheep at 3-month-old. (B) *MSTN* mRNA expression level of gluteus medius in WT (n=3) and *MF*^{+/-} (n=4) sheep. (C-D) *MSTN* protein expression level of gluteus medius in WT (n=3) and *MF*^{+/-} (n=4) sheep. (E) The proportion of centrally nucleated myofibres between WT (n=3) and *MF*^{+/-} (n=4) sheep. (F) The mRNA expression of muscular dystrophy related genes between WT and *MF*^{+/-} sheep (n=4). Data: mean $\pm$ SEM. Unpaired student's t-test was used for statistical analysis. All student's t-test were performed after the equal variance test, otherwise the t-test with Welch's correction were used. **P* < 0.05, ***P* < 0.01, and ****P* < 0.001.

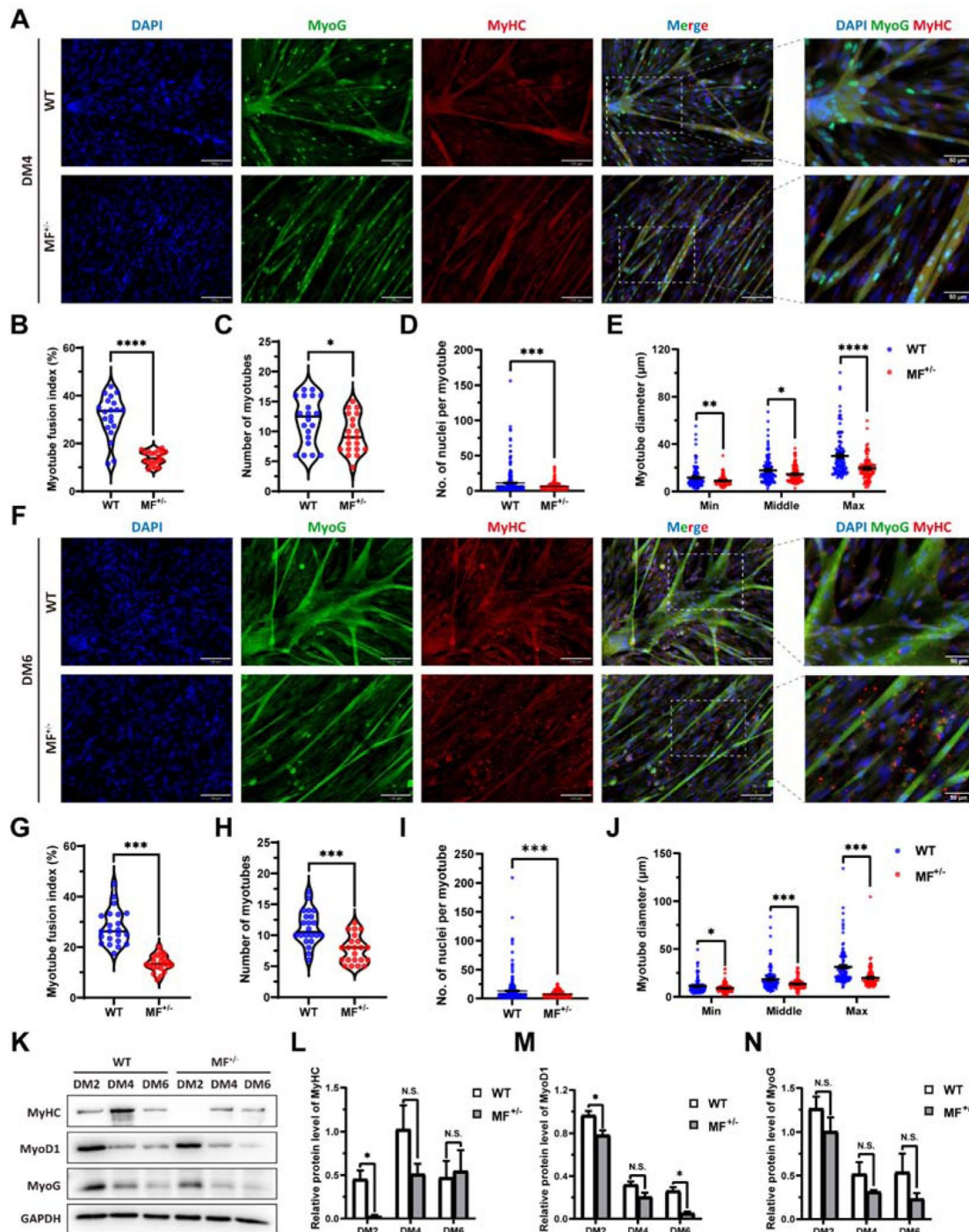


Figure S3

The myogenic differentiation ability of MF^{+/-} cells was continuously inhibited

(A) The MyoG and MyHC immunofluorescence staining of myotubes at DM4. Scale bar 130 μ m. (B-E) The myotube fusion index, number of myotubes, number of nuclei per myotube and the myotube diameter at DM4 (n=3). (F) The MyoG and MyHC immunofluorescence staining of myotubes at DM6. Scale bar 130 μ m. (G-J) The myotube fusion index, number of myotubes, number of nuclei per myotube and the myotube diameter at DM6 (n=3). (K-N) The protein expression levels of myogenic differentiation markers during myogenic differentiation between WT and MF^{+/-} cells (n=2-3). Data: mean $\pm$ SEM. Unpaired student's t-test and chi square test were used for statistical analysis. All student's t-test were performed after the equal variance test, otherwise the t-test with Welch's correction were used. * $P < 0.05$, ** $P < 0.01$, *** $P < 0.001$, and **** $P < 0.0001$.

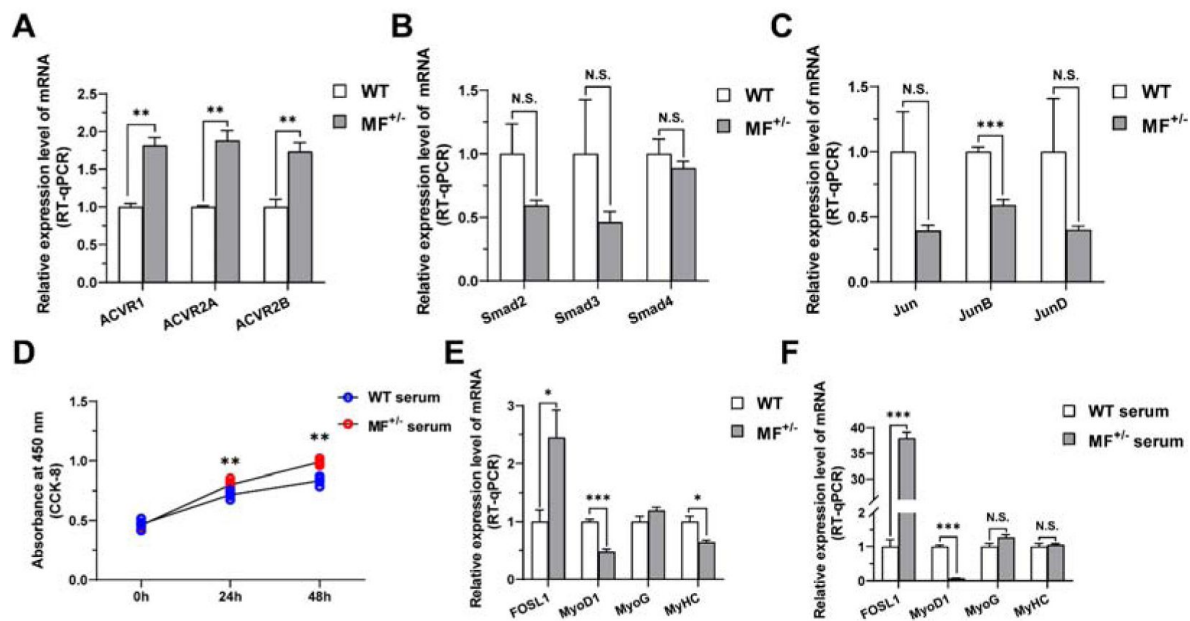


Figure S4

The effects of MSTN signaling pathway and MF^{+/-} serum on proliferation and differentiation of skeletal muscle satellite cells

(A) The mRNA expression levels of type I and type II receptors of MSTN between WT (n=3) and MF^{+/-} (n=4) sheep gluteus medius. (B) The mRNA expression levels of the Smad family downstream of MSTN between WT (n=3) and MF^{+/-} (n=4) sheep gluteus medius. (C) The mRNA expression levels of the Jun family downstream of MSTN between WT (n=3) and MF^{+/-} (n=4) sheep gluteus medius. (D) The number of cells detected by CCK-8 at 0h, 24h, and 48h after culturing cells with serum from WT sheep and MF^{+/-} sheep (n=4-5). (E) The mRNA expression levels of FOSL1 and myogenic differentiation markers between WT and MF^{+/-} sheep skeletal muscle satellite cells cultured and induced differentiation in sheep serum (n=4). (F) The mRNA expression levels of FOSL1 and myogenic differentiation markers in skeletal muscle satellite cells cultured and induced differentiation by serum from WT and MF^{+/-} sheep (n=4). Data: mean ± SEM. Unpaired student's t-test and chi square test were used for statistical analysis. All student's t-test were performed after the equal variance test, otherwise the t-test with Welch's correction were used. *P < 0.05, **P < 0.01, and ***P < 0.001.

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Editors

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Reviewer #3 (Public Review):

Although the authors findings are interesting, they do little to demonstrate new scientific information or advancements in producing genetically modified livestock with improved production characteristics. While the MSTN^{Del273} sheep exhibited an increased number of muscle fibers, the data provided did not demonstrate a significant improvement in meat production, quality or quantity in the MSTN^{Del273} sheep vs WT.

The manuscript is very long, complicated and difficult to read, given the minimum amount of significant information that is provided. It reads more like a graduate student thesis than a scientific manuscript ready for publication. Given the significant findings are so minimal, the amount of text provided, figures and tables are excessive. A large number of different molecular techniques are employed to try and decipher the mechanism(s) that result in the observed phenotype = double muscling. The authors focus on the MEK-ERK-FOSL1 pathway and suggest this is the key pathway/mechanism resulting in the phenotype observed in MSTN^{Del273} sheep. However, they provide very little "significant" evidence to support this. RNA-Seq data demonstrated that hundreds of different genes were either upregulated or down-regulated, but the authors chose to only focus on FOSL1 and associated genes. The findings do not support the idea that FOSL1 is not involved, but neither do they strongly support FOSL1 involvement. The observations made by the authors could be co-incidental and not causative in nature.

The authors indicate that sgRNA design changes in addition to changing the molar ratio of Cas9MRNA:sgRNA improved the ability to generate biallelic homozygous mutant sheep; however, the data provided to not demonstrate any significant difference. Given the small number of sheep that were actually produced and evaluated, it is extremely difficult to demonstrate anything that was analyzed to be significantly (statistically) different between MSTN^{Del273} sheep and WT, yet the authors seem to ignore this in much of their discussion. There is no explanation as to why the authors started with sheep that were FGF5 knockouts. The reviewer assumes that this was simply a line of sheep available from previous studies and the goal was to produce sheep with both improved hair/wool characteristics in addition to improved muscle development. However, the use of FGF5 knockout sheep complicates the ability to accurately decipher the unique aspects associated with targeting only myostatin for knock-out. At minimum, this is a variable that has to be considered in the statistical analysis. No information is provided on the methods used to produce the MSTN^{Del273} sheep, which seems fundamentally important. It is assumed they were produced by injecting one-cell zygotes then transferring these into surrogate females, but given the information provided, it is impossible to know. Certainly, the methods employed could have a profound effect on the outcome. There is no information provided on the sex of the animals produced and then analyzed.

Comments on revised version:

The manuscript by Chen et al. is improved and demonstrates successful gene editing in sheep embryos to obtain biallelic mutation of Mstn and FGF5. Despite the improvements in the revised manuscript, the cellular and molecular mechanism remain inadequate to conclude whether Fosl1 indeed acts downstream of myostatin. In addition, there is little that is new direction versus confirmatory for what is already well know regarding Mstn and FGF5

There are also a number of editorial mistakes e.g. the authors refer to tables S1-S4 in the materials and methods and results section, but there is no table S1-S4 provided.

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Author response:

The following is the authors' response to the original reviews.

eLife assessment

The authors present a useful analysis of the phenotype of sheep in which the muscle developmental regulator myostatin has been mutated in a FGF5 knockout background.

The goal was to produce sheep with a "double-muscled" phenotype, yet the genetically engineered sheep exhibited meat with a smaller cross-sectional area and higher number of muscle fibers. The work extends the extensive body of knowledge already published in this area. The authors provide evidence using in vitro experiments that Fos1 regulates myogenesis, but the strength of evidence relating to the muscle phenotype and underlying cellular and molecular mechanism is inadequate.

Thanks for assessment. According to the reviewers' comments, we have supplemented and updated the data on muscle phenotypes, and the molecular mechanisms also have been supplemented accordingly, such as FOSL1 silencing and inhibition, as well as possible secondary fusion of myoblasts regulated by calcium signaling. Meanwhile, considering the suggestions of editors and reviewers, we have also supplemented the data on serum MSTN regulation. Given that the phenotype of MSTN gene editing is mutation site dependent, we directly cultured skeletal muscle satellite cells using serum from WT and MF+/- sheep, and showed that the serum regulation cannot be ignored after _MSTN_Del273C mutation with FGF5 knockout.

Public Review:

Chen and collaborators first analysed in sheep embryonic gene editing using CRISPR-Cas9 technology to invalidate the two alleles of Mstn and Fgf5 genes by using different ratios of Cas9 mRNA and sgRNA. They showed that a ratio of 1:10 had highest efficiency and they successfully generated two sheep with biallelic mutations of both genes. Materials and Methods on the generation of gene edited sheep is entirely missing. The data on these gene edited sheep have been already published twice by the authors in different contexts. Other groups reported on gene editing of Mstn or Fgf5 in sheep embryos and the resulting phenotypes.

We thank the reviewers for pointing out our negligence and shortcomings. We have provided detailed information on the generation method of gene editing sheep in the Materials and Methods. Briefly, gene-edited sheep were produced by injecting MSTN sgRNA, FGF5 sgRNA, and Cas9 mRNA into embryos in different ratio.

Although the findings are interesting, they do not provide sufficiently new scientific information or advancements in producing genetically modified livestock with improved production characteristics. While the MSTN^{Del273} sheep exhibited an increased number of muscle fibers, the data provided did not demonstrate a significant improvement in meat productions, quality or quantity in the MSTN^{Del273} sheep vs WT.

Thank you very much for your constructive comments. Considering the lack of data on improving production traits, we have further supplemented the data on meat yield and quality of _MSTN_Del273C mutation with FGF5 knockout sheep in Table S6-10. Although these improvements were not significant enough, our data showed increased meat production traits in _MSTN_Del273C mutation with FGF5 knockout sheep, such as the proportion of hind leg meat to carcass and the proportion of gluteus medius to carcass. For example, the proportion of hind leg meat was significantly increased by 21.2% (Table S7), and the proportion of gluteus medius in the carcass of MF+/- sheep was significantly ($P < 0.01$) increased by 26.3% compared to WT sheep (Figure 2K). In addition, there were no significant ($P > 0.05$) differences in pH, color, drip loss, cooking loss, shearing force, and amino acid content of the longissimus dorsi between WT and MF+/- sheep (Table S8-10). All these results demonstrated that the _MSTN_Del273C mutation with FGF5 knockout sheep had well-developed hip muscles with smaller muscle fibers, which do not affect meat quality, and this phenotype may be dominated by MSTN gene.

The authors indicate that sgRNA design changes in addition to changing the molar ratio of Cas9:mRNA:sgRNA improved the ability to generate biallelic homozygous mutant sheep; however, the data provided do not demonstrate any significant difference. Given the small number of sheep that were actually produced and evaluated, it is extremely difficult to demonstrate anything that was analyzed to be significantly (statistically) different between MSTN^{Del273} sheep and WT, yet the authors seem to ignore this in much of their discussion. There is no explanation as to why the authors started with sheep that were FGF5 knockouts. The reviewer assumes that this was simply a line of sheep available from previous studies and the goal was to produce sheep with both improved hair/wool characteristics in addition to improved muscle development. However, the use of FGF5 knockout sheep complicates the ability to accurately decipher the unique aspects associated with targeting only myostatin for knock-out. At minimum, this is a variable that has to be considered in the statistical analysis. No information is provided on the methods used to produce the MSTN^{Del273} sheep, which is fundamentally important. It is assumed they were produced by injecting one-cell zygotes then transferring these into surrogate females. The methods employed might have a profound effect on the outcome.

We greatly appreciate your review. In the current study, we did not discuss the impact of changes in sgRNA design on the ability to generate biallelic homozygous mutant sheep. In fact, we focused on the delivery molar ratio of Cas9 mRNA to sgRNA and found that increasing the molar ratio of Cas9:sgRNA can improve the ability to produce homozygous biallelic mutations in sheep. We apologize for neglecting this statistical analysis, which was tested for significance of differences in the revised version by the chi-square test. Other restrictions related to the actual production and evaluation of the number of sheep were analyzed in our additional discussion. It should be explained to the reviewers that the gene-edited sheep we produced did not start with FGF5 knockout sheep. As hypothesized by the reviewers, we used a one-step method to simultaneously edit the two genes of MSTN and FGF5 to concomitantly increase muscle yield and improve wool characteristics in sheep, which resulted in knockout of the FGF5 gene and mutation of the MSTN gene. As speculated by the reviewers, the MSTN ^{Del273C} mutation with FGF5 knockout sheep was generated by injecting sgRNA and Cas9 mRNA of MSTN and FGF5 into a single fertilized egg and then transplanted into a surrogate mother. We have provided detailed information on the generation method of gene edited sheep in the Materials and Methods section.

Authors genotyped one sheep with a biallelic three base pair deletion in Mstn exon 3 and a compound heterozygote mutation in Fgf5 with a 5 nucleotides deletion on one allele and 37 nucleotides deletion on the other allele, partially spanning over the same region. This sheep developed a double muscle phenotype, which was documented using photography and CT scan. The hair phenotype was not further addressed, but authors referred to a previous publication.

Thank you for your review. In the current study, we only focused our perspective on the muscle phenotype, while the data on the hair phenotype involved another study. Therefore, we referred to our previous publication on hair phenotypes, in which the mutation locus in FGF5 gene-edited sheep is the same as in the current study.

Authors performed morphometric studies on two distinct muscles, longissimus dorsi and gluteus medius, and found a profound fiber hypotrophy in the Mstn^{-/-};Fgf5^{-/-} double mutants, with a shift from larger fiber diameter to smaller fiber sizes. Morphometric studies showed only a low percentage of fibers in wt and mutant sheep had fiber cross sectional areas larger than 800 μm^2 , whereas about 30% in wt and about 60% in the mutant had CSA of <400 μm^2 . The report of one case, without reproducing the

phenotype in other sheep, is scientifically insufficient. The fiber sizes in wt sheep remains far below previously published reports in sheep (about 3-5 times smaller) and as compared to other species, which suggests a methodological error in morphometric methods.

We greatly appreciate your careful review. There is indeed an error in morphological analysis of the MF^{-/-} sheep longissimus dorsi and gluteus medius muscles. After carefully checked, we found that the reason for the fiber sizes in WT sheep remains far below previously published reports in sheep was due to the incorrect use of scale. Thus, we re-scanned the tissue sections and re-calculate the cross-sectional area of muscle fibers and the number of muscle fiber cells per unit area with the correct scale. In this case, the average cross-sectional area of muscle fibers in WT sheep was approximately 1800 μm^2 , which is consistent with the previous report. We once again salute the reviewing expert for such a careful and conscientious review. Considering the profound fiber hypotrophy in MSTN_Del273C mutation with *FGF5* knockout sheep as pointed out by the reviewer, we performed a statistical analysis on the proportion of centrally nucleated myofibres between WT and MF^{+/-} sheep, which can characterize the occurrence of muscle fiber hypotrophy. The results showed that there was no significant difference in the proportion of centrally nucleated myofibres between WT and MF^{+/-} sheep (Figure S2D). At the same time, we also analyzed the mRNA expression levels of muscle fiber hypotrophy and muscle atrophy related genes, such as MTM1, DMD, IGF1, SMN1, and GAA. Although the levels of MTM1, IGF1, SMN1, and GAA were significantly increased (Figure S2E), this elevation did not result in the occurrence of muscle fiber hypotrophy and muscle atrophy, but was beneficial for muscle formation. Therefore, we suggest that the phenomenon produced by MSTN_Del273C mutation with *FGF5* knockout may not be muscle fiber hypotrophy. Because MSTN_Del273C mutation with *FGF5* knockout significantly promotes the proliferation of sheep skeletal muscle satellite cells (Figure 3A-F), and more importantly, its muscle phenotype in MF^{-/-} and MF^{+/-} sheep were improved, including the "double-muscle" phenotype of the rump (Figure 2A), the proportion of gluteus medius in the carcass (Figure 2K), and the proportion of hind leg meat (Table S7).

The authors also investigated the influence of Fgf5 mutation on muscle development. They determined fiber cross sectional area in heterozygous Fgf5 mutant (number of investigated animals not given) and conclude that Mstn mutation but not Fgf5 mutation caused the double muscle phenotype. Results are insufficient to support this conclusion. Firstly, authors investigated heterozygous FGF5 sheep and not homozygous mutants. Secondly, FGF5 has previously been shown to stimulate expansion of connective tissue fibroblasts and to inhibit skeletal muscle development during limb embryonic development (Clase et al. 2000). Of note, Mstn is also expressed during embryonic development. A combined knockout could therefore entail synergistic effects and cause muscle hyperplasia that is not found in individual knockout, a hypothesis that was not addressed by the authors.

Thank you very much for your critical review, which is very valuable for improving the quality of our manuscript. We have given the number of animals studied in all figure legends. Given the lack of MSTN and FGF5 single gene edited sheep, both homozygous and heterozygous sheep, especially MSTN single gene edited sheep, we have weakened the view that MSTN mutations rather than FGF5 mutations lead to "double-muscle" phenotype in conclusion and discussion. As you have mentioned, our current data is indeed insufficient to support this conclusion. In addition, considering the expression of MSTN and FGF5 in embryonic development and their regulation of skeletal muscle development, we examined the expression of MSTN and FGF5 in individual development after MSTN_Del273C mutation with *FGF5* knockout (Figure S2A). However, these results are limited by the animals involved in embryonic development, especially single gene edited embryos. We greatly appreciate

your very meaningful and valuable comments on the possible synergistic effects of combined knockdown. We will prepare MSTN and FGF5 single gene edited sheep to further explore possible synergistic effects in the following study.

The authors generated and studied an F1 generation of mutant sheep with heterozygous mutation in Mstn and Fgf5. In Mstn+/-;Fgf5+/-, gluteus medius muscle was found to be larger compared to wt sheep, whereas other muscles were smaller, and overall meat quantity did not change. Morphometric studies revealed a similar muscle fiber hypotrophy and muscle hyperplasia as in the Mstn-/-;Fgf5-/- gluteus muscle.

Thank you for your comments. We found that the proportion of gluteus medius in MF+/- sheep was larger than that in WT sheep, and in addition, the proportion of hind leg meat also significantly increased (Table S7). Morphological analysis shows that MF+/- sheep exhibited a myofiber hyperplasia phenotype similar to MF-/- sheep.

In the next part of results, authors investigated the presence of myostatin protein in homozygous Mstn muscle using immunohistochemistry and found no differences compared to wt, however, positive and negative controls are missing. The also determined Mstn transcription and protein quantity using WB in heterozygous Mstn muscle and found no difference. The authors did not provide data to explain of why the herein generated Mstn mutation causes muscle fiber hypotrophy, whereas most work on myostatin abrogation demonstrated fiber hypertrophy.

Thank you very much for your constructive comments. Due to the lack of necessary positive and negative controls in immunohistochemistry study, we decided to delete the data on immunohistochemistry in the manuscript to further streamline it. In the current study, although mutations in MSTN lead to a decrease in the cross-sectional area of individual fibers, the number of muscle fibers per unit area were increased, and the final result was an increase in muscle volume and a “double-muscle” phenotype, as well as an increase in the proportion of gluteus medius to carcass (Figure 2K) and the proportion of hind leg meat (Table S7). Importantly, there was no significant difference in the proportion of centrally nucleated myofibres between WT and MF+/- sheep (Figure S2D), and the elevated expression levels of muscle fiber hypotrophy and muscle atrophy marker genes MTM1, IGF1, SMN1, and GAA are more beneficial for muscle health. Therefore, we support that this is not a muscle fiber hypotrophy. As for the phenotype of muscle fiber hypertrophy demonstrated by most myostatin abrogation studies, we analyzed the possible reasons in the discussion, that is, the effect of MSTN mutation on muscle fiber phenotype may be mutant site-dependent.

Authors then isolated myoblasts from hind limbs of 3-month-old sheep fetuses and cultured in presence of 20% fetal bovine serum before switching to differentiation medium containing 2% horse serum. The cultures showed increased proliferation of Mstn+/-;Fgf5+/- myoblasts as well as downregulation of genes associated with muscle differentiation as well as reduced fusion index. No experiments were performed to assure whether the myostatin and FGF5 pathways were inhibited. No control experiments using supplementation with recombinant proteins and using growth factor depleted culture supplements were performed. As FGF5 and myostatin are secreted factors, evidence is missing whether this led to conditioning of the culture medium. Of note, previous work in mice demonstrated that the double muscle phenotype developed independent of satellite cells activity (Amthor et al. 2009).

We greatly appreciate your valuable suggestions. In addition to detecting the MSTN pathway at the cellular level, we also assayed the expression of MSTN receptors and downstream Smad and Jun families in the gluteus medius, and found that _MSTN_Del273C mutation with FGF5 knockout led to upregulation of two receptors, while the expression of downstream

Smad and Jun families was also inhibited to varying degrees (Figure S4A). Considering the possible serum regulation, we also supplemented the data on serum MSTN regulation. Given that the phenotype of MSTN gene editing is mutation site dependent, we directly cultured skeletal muscle satellite cells using serum from WT and MF+/- sheep. We found that serum from MF+/- sheep promoted the proliferation of skeletal muscle satellite cells (Figure S4D). _MSTN_Del273C mutation with *FGF5* knockout promoted FOSL1 expression using WT sheep serum (Figure S4E), which was similar to the results of FBS culture and HS induction. The serum from MF+/- sheep strongly stimulated FOSL1 expression and the inhibition of MyoD1 (Figure S4F). These results indicate that serum regulation cannot be ignored after _MSTN_Del273C mutation with *FGF5* knockout.

Authors then performed RNA seq from Mstn+/-;Fgf5+/- muscle and found a number of differentially expressed genes, but none has been previously reported being involved in the myostatin signaling pathway, so the authors chose to only focus on FOSL1 and associated genes. Authors then demonstrated that Pdpn and Ankrd2 were upregulated during myogenic differentiation, whereas FOSL1 was downregulated. Moreover, Fosl1 transcription was upregulated in myoblasts and myotubes from Mstn+/-;Fgf5+/- muscle. Authors showed an interaction between Fosl1 and Myod1. Moreover, authors demonstrated that Pdl1 directly binds to the Myod1 promoter. Authors also found decreased p38 MAPK protein levels in proliferating myoblasts from Mstn+/-;Fgf5+/- muscle and increased p38 MAPK in differentiating myotubes.

In the revised version, we have streamlined this section by removing content such as PDPN, AKNRD2, and p38 MAPK, aiming to focus on the MEK-ERK-FOSL1 axis. Meanwhile, we further confirmed the regulatory effect of FOSL1 on MyoD1 by dual luciferase assay.

Furthermore, gain-of-function by overexpressing FOSL1 promoted cell proliferation and inhibited differentiation, and tert-butylhydroquinone, an indirect activator of FOSL1 also inhibited myogenic differentiation. The findings do not support the idea that FOSL1 is not involved, but neither do they strongly support the involvement of FOSL1. The observations made by the authors could be co-incidental and not causative in nature.

We greatly appreciate the valuable suggestions provided by the reviewers, which are of great significance for improving our manuscript. Considering the reviewers' suggestions, we supplemented the FOSL1 loss-of-function experiments and found that interfering with FOSL1 can inhibit the proliferation and promote differentiation of skeletal muscle satellite cells, which is contrary to the results of overexpression of FOSL1 (Figure 6). Meanwhile, we also used the inhibitor PB98059 to inhibit the ERK pathway to indirectly inhibit the activity of FOSL1, and the results showed that inhibition of FOSL1 activity also promoted myogenic differentiation (Figure 7F-G). These results could further support the important role of FOSL1.

The manuscript by Chen et al. demonstrated successful gene editing in sheep embryos to obtain biallelic mutation of Mstn and FGF5. The resulting double muscle phenotype resulted from fiber hypotrophy and hyperplasia, which contradicts findings in the literature. Chen et al. generated F1 heterozygous offsprings, in which Mstn transcription and translation did not change. Myoblasts from these animals showed increased proliferation and decreased differentiation, which authors interpreted as the underlying cellular mechanism of the double muscle phenotype. However, no work on muscle development in these animals is presented. Important in vitro control experiments are missing. Chen and collaborators found Fosl1 as a differentially expressed gene in Mstn+/-;Fgf5+/- muscle. Fosl1 drives myoblast proliferation and has direct regulatory effect on the Myod1 promoter. The cellular and molecular mechanism of Fosl1 during myogenesis is novel and solid evidence. However, data remain inadequate to conclude whether Fosl1 indeed acts downstream of myostatin.

We greatly appreciate the reviewers for their insightful insights and very constructive suggestions, which were very helpful for further improving our data. In our study, although the mutation in MSTN resulted in a decrease in the cross-sectional area of individual muscle fibers, the number of muscle fibers per unit area increased, which ultimately resulted in an increase in muscle size and the development of a "double-muscle" phenotype. Therefore, we support that this is not a manifestation of muscle fiber dystrophy, and the detection of some marker genes for muscle fiber dystrophy and the proportion of central nucleus of muscle fibers also support this hypothesis (Figure S2E-F). In addition, the results such as a reduced cross-sectional area of per muscle fibers in our findings contradict the literature on muscle fiber hypertrophy, which may be due to phenotypic differences caused by mutations at different sites of MSTN, and perhaps may also be species-related. For example, the Belgian blue cattle with a natural mutation in the MSTN gene have an increased number of myofibers and a reduced myofiber cross-sectional area [1], and knockdown of the MSTN gene leads to an increase in the cross-sectional area of muscle fibers in mice, without affecting the number of muscle fibers [2,3], as we further described this in discussion. It should be noted that the possible complementary regulation of FGF5 cannot be ruled out either, but unfortunately, this makes the problem extraordinarily complex. We plan to produce single mutant sheep with segregation of the MSTN and FGF5 genes in subsequent studies and give full consideration to the current problem. Regarding the muscle development of gene-edited animals, due to the limitations of large animal conditions and limited editing individuals, we have not comprehensively evaluated the process of muscle development *in vivo* to further improve the potential cellular mechanisms of muscle phenotype, except for evaluating the expression of MSTN and FGF5 at the age of 3 months of individual development and the expression of MSTN at 12 months of age (Figure S2A). To determine whether FOSL1 indeed acts downstream of MSTN, we supplemented the expression levels of FOSL1 under serum regulation to support our conclusions (Figure S4D-F).

[1] Wegner J, Albrecht E, Fiedler I, Teuscher F, Papstein HJ, Ender K. Growth- and breed-related changes of muscle fiber characteristics in cattle[J]. Journal of Animal Science, 2000,78:1485-1496.

[2] Nishi M, Yasue A, Nishimatu S, Nohno T, Yamaoka T, Itakura M, Moriyama K, Ohuchi H, Noji S. A missense mutant myostatin causes hyperplasia without hypertrophy in the mouse muscle[J]. Biochemical and Biophysical Research Communications, 2002,293:247-251.

[3] Zhu X, Hadhazy M, Wehling M, Tidball JG, McNally EM. Dominant negative myostatin produces hypertrophy without hyperplasia in muscle[J]. FEBS Letters, 2000,474:71-75.

As the significant findings are minimal, the amount of text provided, figures and tables are disproportionally excessive. A large number of different molecular techniques are employed to try and decipher the mechanism(s) that result in the observed phenotype = double muscling. The authors focus on the MEK-ERK-FOSL1 pathway and suggest this the key pathway/mechanism resulting in the phenotype observed in MSTN^{Del273}sheep. However, they provide very little solid evidence to support this notion.

Thank you for your review. We have substantially streamlined the manuscript, removed some irrelevant information, and provided all unnecessary figures and tables as supplementary information. Meanwhile, we have added new data to further support that _MSTN_DelC273 mutation generates a muscle phenotype through the MEK-ERK-FOSL1 pathway.

The manuscript is very long, complicated and difficult to read, given the minimum amount of significant information that is provided. It requires major rewriting to be published. Further, it misses information in material methods, on the generation of

animals, on histological techniques and morphometric studies. There is no information provided on the sex of the animals produced and then analyzed. There are also a number of editorial mistakes e.g. the authors refer to tables S1-S4 in the materials and methods and results section, but and there is no table S1-S4 provided.

Thank you for your review. We have greatly streamlined and significantly revised the manuscript. At the same time, we have supplemented detailed information on animal generation, histologic and morphological studies in materials and methods, as well as the information on gene-edited animal production, including gender, age, and so on. Finally, we reviewed the entire manuscript and updated any possible omissions or negligence, such as those oversights like tables S1-S4.

Recommendations for the authors:

Suggestions to improve the paper (see also public review):

- Include the method part of generating the gene edited animals.

We thank the editor and reviewers for pointing out our negligence. We have provided detailed information on the generation method of gene-edited sheep in Materials and Methods, which was produced by injecting MSTN sgRNA, FGF5 sgRNA, and Cas9 mRNA into embryos in different ratios.

- Increase number of Mstn^{-/-};Fgf5^{-/-} experimental animals allowing for acquisition of statistically relevant data. This is very important as the muscle phenotype of the F1 generation is not obvious. Authors should provide data that the Mstn mutation indeed invalidates myostatin signaling. They should provide data on myostatin protein Mstn transcription as well on myostatin target genes in Mstn^{-/-};Fgf5^{-/-} sheep.

Many thanks to the editor and reviewers for their constructive suggestions. The strategy of using MF^{-/-} sheep to validate the transcription and target gene data of myostatin is indeed the best. However, we only generated one MF^{-/-} sheep, which seriously limits the implementation of such an optimal strategy and may also make statistical analysis based on MF^{-/-} sheep unreliable. Considering these factors, our current study mainly focuses on heterozygous MF^{+/-} sheep. We are planning to generate single gene homozygous mutant sheep for MSTN and FGF5 gene separation in subsequent studies and to give full consideration to the current issue.

- They should also provide data on myostatin target genes in muscles from heterozygous animals.

Thank you for your very informative suggestions. We have quantitatively detected the mRNA expression levels of the receptors and downstream target genes of MSTN in the gluteus medius of heterozygous MF^{+/-} sheep. Compared with WT sheep, the mRNA expression levels of type I receptor (ACVR1) and type II receptor (ACVR2A, ACVR2B) were highly significantly increased in the muscle of MF^{+/-} sheep (Figure S4A), there was no significant change in mRNA expression levels in the Smand family (Figure S4B), whereas the mRNA expression levels of JunB of Jun family, a downstream target gene of MSTN, were significantly down regulated (Figure S4C). These results suggest that the effect of _MSTN_Del273C with FGF5 knockout may not be limited to MEK-ERK-FOSL1. Again, we would like to thank the editor and reviewers for their constructive suggestions, which provide a new direction for us to further deepen our insight into the mutations of MSTN gene.

- The morphometric results on fiber CSA seem wrong. By looking at the fiber sizes and size bar in Figure 2 H would bring to far higher estimated CSA. There must be a

systematic error in using the morphometric algorithm.

Thank you very much for your careful review. There were indeed some errors in morphological analysis of the MF^{-/-} sheep longissimus dorsi and gluteus medius. After checking, we found that the reason why the muscle fiber size was much lower than the data in the previously published sheep report was due to the incorrect use of scale bar. To this end, we re-scanned the tissue slices and used the correct scale bar to re-counted the cross-sectional area of muscle fibers and the number of muscle fiber cells per unit area. In this case, the average cross sectional area of muscle fibers in WT sheep was similar to the previous report.

- The labeling of the ordinate of Fig. 2I is not readable ($\times 1000 \mu\text{m}^2$, or $\times 100 \mu\text{m}^2$?). Authors should make sure that they look at the same muscle part, as fiber sizes can highly vary depending on exact anatomical situation. In small laboratory animals, entire muscle cross sections are usually analyzed to prevent such bias. This may proof difficult in large animals, however, small muscles could easily be identified and cross sections of entire muscles be analyzed. As myostatin KO concerns all skeletal muscles, authors could consider muscle such as FDB or extraocular muscles.

Thank you for your careful review and suggestions. The vertical axis of Figure 2I is in the units of $\times 1000 \mu\text{m}^2$, and each data point represents the actual measured area of each muscle fiber. Because there are significant differences in muscle fiber size, we visualized the measurement values of all individual muscle fiber areas, and the average value of the scatter plot was used as the average area of all muscle fibers. We did this to provide a more intuitively display the distribution of muscle fiber size.

- The material of methods of muscle histology and morphometric studies must be included.

Thank you for your suggestions. We have supplemented the methods of muscle histology and morphology study, as well as statistical methods for cross-sectional area and quantity of muscle fibers in the material methods.

- In figures, numbers of experimental animals be given throughout, as well as number of technical repeats. The authors need to provide some minimal data on how the genetically engineered sheep were produced, in addition to how many, the sex etc.....and which of these were analyzed to obtain the data. It is impossible to know when reading this manuscript whether data involving, for example gene seq, westerns, microscopic images etc involves one sheep or some compilation of data.

Thank you very much for your constructive suggestions, which is of great guiding significance for improving the quality of our manuscript. We have clearly stated the number of experimental animals and the number of any biological replicates in all figure legends. Meanwhile, we have provided detailed information on the generation method of gene edited sheep in the Materials and Methods, which was produced by injecting MSTN sgRNA, FGF5 sgRNA, and Cas9 mRNA into embryos in different ratios.

- As authors work on Mstn;Fgf5 double KO animals, they should explore whether Fgf5 is expressed in developing sheep muscle, and whether combined KO entails a synergistic effect on muscle development.

We detected the expression of FGF5 in muscle tissue of WT and MF^{+/-} sheep at 3 months of age of individual development, which was significantly reduced compared to WT sheep (Figure S2A). We greatly appreciate your very meaningful and valuable comments on the

possible synergistic effects of combined knockdown. Due to the limitations of single gene knockout of MSTN and FGF5 in sheep in our current study, especially their homozygous mutants. We will prepare MSTN and FGF5 single gene edited sheep to further explore possible synergistic effects in the following study.

- The authors should address the question of why their mstn mutation causes fiber hypotrophy, whereas most other work reported the opposite. Why would herein generated mutation act differently? Does mutated myostatin gain a different biological effect? Does it bind to different receptors?

Thank you very much for your valuable comment. Regarding the possibility of muscle fiber dystrophy in _MSTN_Del273C mutation with *FGF5* knockout sheep, we have performed a statistical analysis of the proportion of central nucleus of muscle fibers in MF+/- sheep, which can characterize the occurrence of muscle dystrophy in some extent. The results showed that there was no significant difference in the proportion of central nucleus of muscle fibers between WT and MF+/- sheep (Figure S2E). At the same time, we also analyzed the mRNA expression levels of genes MTM1, DMD, IGF1, SMN1, and GAA related to muscle fiber dystrophy and muscle atrophy. Although the levels of MTM1, IGF1, SMN1, and GAA were significantly increased (Figure S2F), this elevation did not lead to the occurrence of muscle fiber dystrophy and muscle atrophy, but instead, it was beneficial for muscle formation. Therefore, we suggested that this phenomenon produced by _MSTN_Del273C mutation with *FGF5* knockout may not be muscle fiber dystrophy, as _MSTN_Del273C mutation with *FGF5* knockout significantly promoted the proliferation of sheep skeletal muscle satellite cells (Figure 3A-F). More importantly, _MSTN_Del273C mutation with *FGF5* knockout improves the muscle phenotype of sheep, including the "double-muscle" phenotype of the rump (Figure 2A), the proportion of gluteus medius to the carcass (Figure 2K), and the proportion of hind leg meat (Table S7). In addition, we analyzed in discussion why the current mutation produces a phenotype different from other work reports, which we suggested that this may be due to different mutation sites. We provided a detailed analysis of this in discussion. It is indeed a very thought-provoking question about whether mutated myostatin acquire different biological effects and whether they bind to different receptors, which we plan to further reveal this in the homozygous MSTN and FGF5 mutant sheep.

- Concerning the in vitro work, authors need to demonstrate whether Mstn and/or FGF5 signaling pathways are altered in myoblasts/myotubes. As both are secreted factors, authors need to show that serum conditioning is changing in myoblast cultures. Authors should perform cultures in which these factors are entirely suppressed and thus signaling pathway shut down. They could use growth factor depleted supplements and/or add myostatin and FGF5 inhibitors to the serum. The need to determine first the individual effect of myostatin and FGF5 and then challenge the combined effect. They also should perform the inverse experiment and supplement cultures with recombinant factors, both as individual approach and combined approach.

We greatly appreciate your valuable suggestions. In addition to detecting the MSTN pathway at the cellular level, we also assayed the expression of MSTN receptors and downstream Smad and Jun families in the gluteus medius, and found that _MSTN_Del273C mutation with *FGF5* knockout led to upregulation of two receptors, while the expression of downstream Smad and Jun families was also inhibited to varying degrees (Figure S4A). Considering the possible serum regulation, we also supplemented the data on serum MSTN regulation. Because we have previously tested inhibitors of MSTN and FGF5, but did not observe any effect, we suggest this may be due to the nonspecificity of the inhibitors, as there are no sheep specific MSTN and FGF5 inhibitors. Given that the phenotype of MSTN gene editing is mutation site dependent, we directly cultured skeletal muscle satellite cells using serum from WT and MF+/- sheep. We found that serum from MF+/- sheep promoted the proliferation of

skeletal muscle satellite cells (Figure S4D). *_MSTN_Del273C* mutation with *FGF5* knockout promoted FOSL1 expression using WT sheep serum (Figure S4E), which was similar to the results of FBS culture and HS induction. The serum from MF+/- sheep strongly stimulated FOSL1 expression and the inhibition of MyoD1 (Figure S4F). These results indicate that serum regulation cannot be ignored after *_MSTN_Del273C* mutation with *FGF5* knockout.

- With above suggested additional experiments, authors would also be able to demonstrate, whether Fos1 is indeed triggered in response to myostatin and/or FGF5 signaling.

To determine whether FOSL1 indeed acts downstream of MSTN, we supplemented the expression levels of FOSL1 under serum regulation to support our conclusions. We found that the serum from MF+/- sheep strongly stimulated FOSL1 expression and the inhibition of MyoD1 (Figure S4F).

- Authors used t-test despite in several tests despite low sample number, which violates as such the assumption of equal variance. Non-parametric tests should be used in this case.

Thank you very much for your valuable comments. We apologize for the previous incorrect use of statistical methods. In the revised version, we have re-analyzed all data. Before performing student's t-test, we first evaluated the assumptions of normal distribution and equal variance. Two-tailed student's t-tests were used only for data that conformed to normal distribution and homogeneity of variance, otherwise corrected Welch's t-tests were performed.

- Authors should state in the legends which statistical test was used.

Thank you for your suggestion. We have clearly stated the statistical testing method used in all figure legends, which is indeed necessary and important.

In general, this manuscript should be dramatically scaled back in terms of content, eliminating unnecessary text, figures and tables that do not play a significant role in the findings that were significant. There is some interesting information and data here that can add to the overall base of knowledge surrounding the production of genetically engineered livestock in which myostatin has been targeted for mutation. However, the authors need to focus on their findings that were significant and strongly supported by the data and statistical analysis. Some discussion of findings that support their ideas/hypothesis, but are not statistically significant is fine. But it should not make up the majority of the manuscript which is the case here.

Thank you for your valuable suggestions, which are essential for improving the quality of our manuscript. We have greatly streamlined and significantly revised the manuscript, removed unnecessary text, figures, and tables.

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